Cost-Sensitive Attribute Reduction for Label Distribution Learning Using Granular-Ball Computing with Objectwise Reconstruction Bounds

Chaoyang Li[a], Xiaohan Li[b], Zhaowen Li[c,*]

[a] *Guangdong University of Science and Technology, Dongguan, 523083, Guangdong, China*
[b] *Faculty of Innovation Engineering, Macau University of Science and Technology, Macau, China*
[c] *Fujian Key Laboratory of Financial Information Processing, Putian University, Putian, 351100, Fujian, China*

Abstract

Aggregate reconstruction criteria do not explicitly enforce fidelity for individual objects: a small mean error can coexist with increased reconstruction error for particular objects. This study formulates cost-sensitive attribute reduction for label distribution learning under fixed-reference objectwise bounds. Each training object's squared reconstruction error is constrained by its full-attribute reference error plus a prescribed tolerance. GBIP-O1 combines label-information geometry and granular-ball reconstruction with complete cost-ordered deletion sweeps. The algorithm preserves feasibility and terminates at an empty subset or a feasible subset that is irreducible under single-attribute deletion. A label-margin condition provides a sufficient guarantee for preserving a unique dominant label. Evaluation on 25 tasks spanning 16 feature families and a separate SCUT-FBP5500 image pipeline examines acquisition cost, reconstruction fidelity, prediction and resource use. The results show that mean preservation can permit individual deterioration, whereas strict objectwise preservation can lead the deletion search to retain all attributes. Matched-constraint comparisons and sensitivity analyses characterize the effects of search strategy, geometry, sample size,

*Corresponding author.
Email addresses: lichaoyang1@gdust.edu.cn (Chaoyang Li), xiaohanl06@student.must.edu.mo (Xiaohan Li), lzwli@ptu.edu.cn (Zhaowen Li)

prices and tolerance. The resulting guarantees apply to the specified training reconstruction constraints; held-out prediction and end-to-end resource use require separate evaluation.

Keywords: Label distribution learning, cost-sensitive attribute reduction, objectwise reconstruction, granular-ball computing, directed dissimilarity

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1. Introduction

Label distribution learning (LDL) represents an instance by a distribution over a set of labels, capturing differences in relative label relevance rather than assigning a single label [1]. While reducing the attributes used to represent an object can lower acquisition and storage costs [2, 3], it may alter the reconstructed label distribution. Traditional aggregate reconstruction criteria allow improvements in some instances to compensate for severe deterioration in others. Consequently, a reduced attribute set yielding a small average error may still substantially distort particular objects. This distinction motivates a reduction objective: minimizing attribute acquisition cost while requiring every object in a training collection to satisfy an explicit reconstruction bound relative to its full-attribute reference.

Facial attractiveness assessment illustrates the distinction between aggregate accuracy and individual fidelity. Rating distributions capture both the relative frequencies of aesthetic assessments and the inherent disagreement among observers—nuances that a single mean score often obscures [4, 5]. When stored facial descriptors are reduced, a small average reconstruction error across a dataset does not preclude a change in the rating distribution of a specific image. Protocols requiring reconstruction fidelity for each stored instance therefore require objectwise verification.

Individual-fidelity requirements may also arise in medical imaging, credit assessment and industrial inspection when an evaluation protocol specifies allowable reconstruction changes for individual records. Emerging literature in medical-imaging machine learning highlights how aggregate performance metrics can conceal important failures on specific data subsets [6]. If an evaluation protocol specifies an allowable reconstruction deviation for each record, aggregate performance alone cannot verify compliance. Establishing a fixed reference and an explicit objectwise bound makes this requirement explicit and auditable: an evaluator can transparently audit the baseline, permitted deviation, and measured change for every inspected record [7].

Its practical significance, however, hinges on how well the chosen reference, reconstruction loss, and tolerance align with downstream task requirements.

The relevance of an objectwise reconstruction bound is further contingent upon the margin of separation between label distribution degrees. When the two largest label degrees are close in magnitude, even a minor perturbation can alter the dominant label, implying that a reconstruction-error bound does not inherently guarantee label or ranking stability [8]. To bridge this gap, we relate the bound directly to the object's label margin, establishing a formal sufficient condition to preserve its unique dominant label. This theoretical guarantee is complemented by empirical assessments of dominant-label, top-two, and within-object ranking shifts, distinguishing the properties certified by the reconstruction constraint from those necessitating independent evaluation.

Existing LDL feature-selection methods predominantly evaluate attribute subsets through dependency, local consistency, or conditional discrimination. Section 2 reviews these approaches alongside information-based selection, granular-ball representations, and acquisition-cost models. GBFRS combines granular-ball fuzzy dependency with local consistency [9], while IWDLFS and IWCLFS employ dependence and conditional discrimination criteria within an informational weighted neighborhood rough-set model [10]. These formulations assess the discriminative structure retained by an attribute subset. In contrast, our formulation addresses a different requirement: whether every training instance's reconstruction error remains bounded by its full-attribute reference error plus a prescribed tolerance. These objectwise bounds define the feasible family, within which GBIP-O1 seeks lower-cost subsets through complete deletion sweeps. Its termination certificate establishes single-deletion irreducibility rather than global cost optimality.

To implement this formulation, we integrate label-information weights and directed dissimilarity into granular-ball reconstruction. Beginning from the full attribute set, the GBIP-O1 algorithm systematically evaluates attribute deletions in descending order of cost, accepting a deletion only when all objectwise bounds remain satisfied. Because removing attributes regenerates the granular balls, distance monotonicity does not guarantee the monotonicity of reconstruction feasibility. Consequently, rejected deletions are revisited across successive complete sweeps until an entire sweep yields no accepted deletions. The algorithm ultimately terminates at a feasible subset that is single-deletion irreducible.

This study contributes to the literature in three respects.

1. A cost-sensitive LDL attribute-reduction framework is formulated under fixed-reference objectwise reconstruction bounds, distinguishing individual feasibility from aggregate quality and defining acquisition-cost minimization over the feasible family.

2. GBIP-O1 integrates label-information geometry, granular-ball reconstruction and complete cost-ordered deletion sweeps. The analysis establishes feasibility preservation, finite termination and single-deletion irreducibility, together with a sufficient label-margin condition for preserving a unique dominant label.

3. Evaluation on 25 tasks spanning 16 feature families and a separate image pipeline characterizes cost–preservation trade-offs through matched-constraint comparisons, geometry ablations and sensitivity analyses. Decision diagnostics and resource measurements distinguish reconstruction guarantees from predictive and computational outcomes.

Study Overview. Figure 1 illustrates the overarching research workflow, tracing the progression from the preservation objective and data protocol to model construction and empirical assessment. The proposed method and comparative selectors utilize designated training partitions, with their respective feature subsets frozen prior to held-out evaluation. Baseline parameters are tuned via the training-internal cross-validation protocol detailed in Section 7. Training reconstruction, held-out prediction, and image-pipeline resource measurements are treated as distinct outcomes. Furthermore, sensitivity analyses examine data geometry, sample size, acquisition costs, and reconstruction tolerance. This study-level schematic complements the fine-grained GBIP-O1 control flow in Figure 3; it introduces no additional experiments and strictly avoids utilizing test labels for geometry fitting or attribute deletion acceptance.

The comparator branch includes GBFRS-EQ, IWDLFS, IWCLFS and GBFRS+O1, with Mean, Full and Empty as reference comparisons. Evaluation covers theoretical guarantees, training feasibility, acquisition cost, label decisions, held-out Brier error, sensitivity, descriptor storage and measured query time. The prediction transformations are fitted on training data before being frozen.

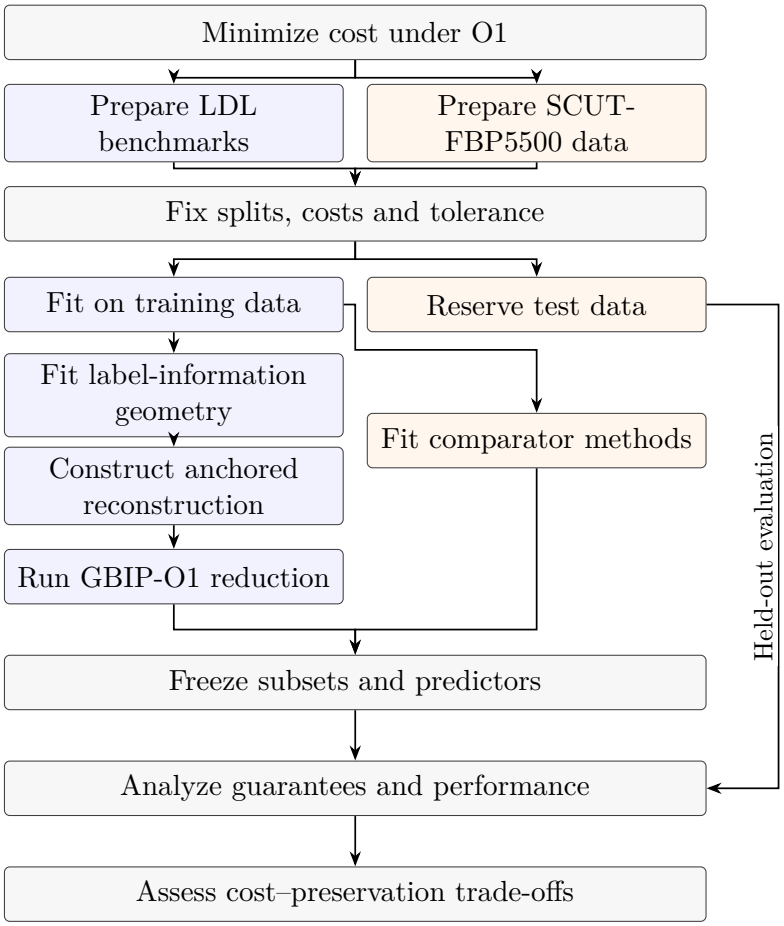


Figure 1: Research Workflow

2. Related Work

2.1. Feature Relevance and Distribution-Valued Supervision

Feature selection fundamentally decouples the choice of variables from the construction of a predictor. A clear distinction between feature ranking, subset evaluation, and search strategies remains central to interpreting any selected subset [11]. Information-theoretic methods constitute a prominent family of relevance criteria: for instance, minimum-redundancy maximum-relevance selection balances feature relevance to the target against inter-feature dependence [12], whereas conditional-likelihood analysis connects various information-based heuristics to approximations of a unified statistical objective [13]. These approaches collectively highlight the necessity of examining what a selection criterion actually preserves. Specifically, a generic relevance score, a downstream predictive loss, and a constraint governing individual training instances represent fundamentally disparate quantities.

In LDL, supervision comprises relative label degrees rather than merely a binary set of relevant labels [1]. Label enhancement addresses the related problem of recovering full distributions from logical labels by exploiting feature-space structure and label correlations [14]. This distinction is crucial when comparing feature-selection methods: relying on an observed distribution as direct supervision versus constructing a distribution from multi-label annotations entails fundamentally different underlying assumptions. Consequently, our reduction framework treats the supplied training distributions as fixed; it neither estimates missing label degrees nor validates whether an enhanced distribution serves as an appropriate reference for a downstream application.

2.2. Feature Selection for Label Distribution Learning

Several lines of research have adapted feature evaluation frameworks to distribution-valued labels. For instance, Qian et al. incorporate feature similarity and label correlation into LDL feature selection [15], Xiong et al. combine label distributions with fuzzy mutual information [16], and Deng et al. evaluate features using dual-similarity neighborhood fuzzy entropy [17]. More recently, You et al. construct fuzzy similarity via the statistical distribution of feature values to evaluate subsets using fuzzy mutual information [18]. Collectively, these formulations successfully retain granular information regarding label relevance—nuances that would otherwise be lost if all associated labels were treated uniformly.

Neighborhood and fuzzy rough-set models offer alternative perspectives for characterizing feature significance. Specifically, Wu et al. establish neighborhood relations within the label-distribution space and evaluate dependency using label weights [19], whereas Deng et al. develop a neighborhood fuzzy rough-set model incorporating the distribution of label significance [20]. Furthermore, Lu et al. combine hierarchical structures with neighborhood granularity to generate feature rankings [21]. Moving beyond a single common feature view, Lin et al. learn feature weights through an LDL prediction model regularized by label correlation [22], while Shu et al. explicitly account for label-specific features [23]. Collectively, these contributions expand the representation and supervision paradigms for feature selection; however, they optimize distinct criteria and should not be viewed as interchangeable implementations of a singular objective.

Furthermore, annotation completeness and feature availability give rise to distinct problem settings. For instance, Qian et al. address incomplete label distributions using a neighborhood-tolerance discrimination index [24], whereas Chen et al. examine streaming features via a dynamic online method driven by label importance and correlation [25]. Subsequently, Qian et al. integrate granular-ball computing with fuzzy rough sets for semi-supervised feature selection under incomplete label distributions [26]. In contrast, our setting fixes both the training collection and the available attributes prior to reduction. Handling missing supervision or incoming features would necessitate additional specifications detailing how the full-attribute reference is constructed or dynamically updated.

The closest experimental comparators in this study are GBFRS, which integrates granular-ball fuzzy dependency with local consistency [9], and I-WDLFS/IWCLFS, which employ dependence and conditional discrimination within an informational weighted neighborhood rough-set model [10]. We strictly distinguish their original selection criteria from the O1-filtered prefix comparators evaluated herein. Meanwhile, the remaining studies reviewed above serve to establish broader methodological context rather than acting as additional empirical baselines. the primary contribution of GBIP-O1 lies in its explicit fixed-reference feasibility requirement and certified deletion endpoint, rather than a generalized claim of predictive superiority over all existing LDL feature-selection paradigms.

2.3. Granular-Ball Representation and Rough-Set Reduction

Rough sets characterize the information retained by an attribute subset through approximation and discernibility [27], with neighborhood formulations extending this perspective to heterogeneous feature spaces [28]. Granular-ball learning complements this by replacing uniform object-level representations with adaptive groupings. For instance, Xia et al. develop efficient and adaptive granular-ball generation methods [29], GBRS connects granular-ball learning with both Pawlak and neighborhood rough-set models [30], and GBG++ further investigates the computational efficiency and stability of ball generation [31]. While these studies establish robust representation and construction mechanisms, their conventional classification or rough-set objectives do not by themselves certify or fulfill the explicit reconstruction constraints defined in our framework.

In LDL, regenerated granular balls jointly influence the neighborhood structure and label-distribution reconstruction. Consequently, the mere geometric fact that removing nonnegative distance terms cannot increase the corresponding directed dissimilarity does not guarantee the monotonicity of the reconstruction criterion following regrouping. This fundamental limitation motivates the explicit feasibility evaluations and iterative complete sweeps implemented in GBIP-O1. The analysis establishes that its termination certificate strictly pertains to the local irreducibility of the final subset under single-attribute deletions; it does not establish global cost minimization, the uniqueness of a reduct, or invariant endpoints across alternative ball-generation procedures.

2.4. Acquisition Cost and the Preservation Requirement

The cost of observing a feature is fundamentally distinct from the cost incurred by a prediction error [2]. Specifically, test-cost-sensitive attribute reduction explicitly incorporates acquisition costs into attributes [32], whereas test-cost-constrained selection restricts the resources available for acquiring informative features [3]. Furthermore, minimum-cost reduction within decision-theoretic rough set models integrates decision costs directly into its reduction objective [33], and cost-sensitive random-forest selection offers another paradigm for selecting informative features while accounting for their measurement costs [34]. Collectively, these studies underscore the imperative of precisely specifying both the resource being minimized and the specific information or performance guarantees that must be preserved.

In our formulation, the objective function corresponds to the sum of selected attribute acquisition costs, whereas feasibility is governed by a distinct reconstruction-error inequality for each training instance relative to a fixed full-attribute reference. Consequently, aggregate relevance or error criteria cannot serve as substitutes for verifying these individual inequalities. Conversely, satisfying these objectwise bounds does not inherently guarantee lower held-out prediction loss or reduced total system latency. To account for this, the image-pipeline experiments report descriptor storage and end-to-end processing timing as separate performance metrics. Furthermore, our framework deliberately defers the scientific choice of the reference and tolerance parameters to the specific application protocol, with our experiments investigating the downstream consequences of these parameter choices rather than advocating for a universal acceptance threshold.

3. Preliminaries

3.1. Test-Cost-Sensitive Label Distribution Information System

Let $O = \{o_1, \ldots, o_n\}$ be a finite, non-empty set of objects, $A = \{a_1, \ldots, a_m\}$ a finite, non-empty set of conditional attributes, and $\mathcal{Y} = \{y_1, \ldots, y_q\}$ a finite, non-empty set of labels. Each attribute $a \in A$ induces a mapping $a : O \to V_a$, where V_a denotes the domain of attribute a. To handle hybrid feature spaces, we partition A as

$$A = A_n \cup A_c, \qquad A_n \cap A_c = \varnothing,$$

where A_n and A_c designate numerical and categorical attribute subsets, respectively. Either subset may be empty.

Definition 1 (Label Distribution Information System)**.** *A label distribution information system is a tuple*

$$\mathcal{S}_L = (O, A, \mathcal{Y}, L),$$

where $L = (\ell_i^k)_{n\times q}$ is the label distribution matrix satisfying the simplex constraints:

$$\ell_i^k \in [0,1], \qquad \sum_{k=1}^{q} \ell_i^k = 1 \quad (i = 1, \ldots, n;\ k = 1, \ldots, q). \tag{3.1}$$

The label distribution associated with object o_i is represented by the vector

$$\boldsymbol{\ell}(o_i) = (\ell_i^1, \ldots, \ell_i^q), \tag{3.2}$$

where ℓ_i^k quantifies the description degree of label y_k for object o_i [1].

In our framework, the complete distribution vector $\boldsymbol{\ell}(o_i)$ serves directly as the supervisory decision signal. Rather than collapsing this vector to its most dominant label, the formulation explicitly preserves fine-grained relative importances. Notably, these description degrees represent quantitative supervisory weights rather than posterior class probabilities.

The proposed information system builds upon Pawlak's classical rough set model [27], incorporating neighborhood-based extensions specifically designed for hybrid data types [28, 35, 36]. While traditional attribute reduction techniques rely on heuristic searches guided by dependency or information measures over crisp equivalence classes [37, 38, 39, 40], our model fundamentally departs from them in two key respects: its decision representation is distribution-valued, and its optimization objective explicitly integrates attribute acquisition costs.

Definition 2 (Label-Induced Fuzzy Decision Sets)**.** *For each label $y_k \in \mathcal{Y}$, define a fuzzy decision set D_k on O via the membership function*

$$\mu_{D_k}(o_i) = \ell_i^k \quad (i = 1, \ldots, n;\ k = 1, \ldots, q). \tag{3.3}$$

The collection $\mathcal{D}_L = \{D_1, \ldots, D_q\}$ constitutes the family of label-induced fuzzy decision sets determined by L.

According to Eq. (3.1), the membership functions satisfy

$$\sum_{k=1}^{q} \mu_{D_k}(o_i) = 1 \quad \text{for all } o_i \in O. \tag{3.4}$$

Consequently, $\mathcal{D}_L$ forms a fuzzy partition-of-unity over O. Unlike traditional decision tables, $\mathcal{D}_L$ does not induce mutually disjoint crisp equivalence classes; instead, an object o_i may simultaneously hold positive membership degrees across multiple fuzzy decision sets D_k. Therefore, theoretical properties established for classical crisp partitions cannot be assumed a priori and must be formally re-established within this soft decision space.

To support the label-weighted statistics introduced in subsequent sections, the total description mass of label y_k is defined as

$$M_k = \sum_{i=1}^{n} \ell_i^k. \tag{3.5}$$

Any statistical quantity containing M_k in the denominator is well-defined if and only if $M_k > 0$. A label satisfying $M_k = 0$ carries zero description mass across all sample instances and is thus naturally excluded from label-weighted computations.

Definition 3 (Test-Cost-Sensitive Label Distribution Information System)**.** *A test-cost-sensitive label distribution information system is defined as*

$$\mathcal{S}_{L,c} = (O, A, \mathcal{Y}, L, c),$$

where $c : A \to \mathbb{R}_{>0}$ assigns a positive acquisition cost to each attribute $a \in A$. Assuming additivity, the total cost of acquiring an attribute subset $P \subseteq A$ is

$$c(P) = \sum_{a \in P} c(a), \qquad \text{with } c(\varnothing) = 0. \tag{3.6}$$

The strict positivity constraint $c(a) > 0$ guarantees strict monotonicity of acquisition costs under subset inclusion—that is, $c(P) < c(Q)$ whenever $P \subsetneq Q$—without imposing any structural monotonicity assumptions on the granular dependency function itself.

Proposition 1 (Crisp Decision Representation as a Special Case)**.** *Suppose each row of L is a one-hot vector (i.e., $\ell_i^k \in \{0, 1\}$ for all i, k). Define a discrete decision mapping $d : O \to \mathcal{Y}$ such that $d(o_i) = y_k$ if and only if $\ell_i^k = 1$. Then, each D_k collapses to the indicator function of the crisp decision class*

$$C_k = \{o_i \in O : d(o_i) = y_k\}.$$

Upon removing empty sets, $\{C_k\}$ forms a strict crisp partition of O.

Proof. For any $o_i \in O$, there exists a unique index k such that $\ell_i^k = 1$. Thus, o_i belongs to exactly one decision class C_k, establishing that $\bigcup_{k=1}^{q} C_k = O$ and $C_k \cap C_h = \varnothing$ for all $k \neq h$. By Eq. (3.3), the membership function reduces to

$$\mu_{D_k}(o_i) = \begin{cases} 1, & \text{if } o_i \in C_k, \\ 0, & \text{if } o_i \notin C_k, \end{cases}$$

which matches the indicator representation of the crisp set C_k. □

Proposition 1 pertains exclusively to the decision representation mapping. Whether the corresponding approximation operators and dependency metrics collapse to classical crisp rough set formulations remains a distinct theoretical question, which must be formally verified once these operators are explicitly defined. Consequently, no automatic inheritance of classical properties—such as monotonic dependency—is assumed herein.

4. Label-Distribution-Based Information Weights and Directed Dissimilarity

4.1. Local KL-Divergence Weights and Directed Dissimilarity

While preserving the source-dependent, fixed-normalization directed aggregation structure, we propose a unified information-theoretic statistic applicable to both numerical and categorical attributes. This statistic quantifies the divergence between an attribute-induced local label profile and the global label profile. Such a modeling choice reflects an information-theoretic design rather than an empirical claim of absolute predictive optimality.

All statistical properties are evaluated exclusively on the fixed training object set O, remaining invariant when comparing alternative attribute subsets $P, Q \subseteq A$. Validation and test labels are strictly excluded from these computations.

4.1.1. Global and Attribute-Induced Local Label Profiles

We define the global mean description degree for label y_k across training instances as

$$\bar{\ell}^k = \frac{1}{n}\sum_{i=1}^{n} \ell_i^k, \qquad \mathcal{K}_+ = \{k \in \{1,\dots,q\} : \bar{\ell}^k > 0\}. \tag{4.1}$$

The vector $\bar{\boldsymbol{\ell}} = (\bar{\ell}^1, \dots, \bar{\ell}^q)$ represents the global label distribution. By non-negativity constraint (3.1), $\bar{\ell}^k = 0$ implies $\ell_i^k = 0$ for all training instances $o_i \in O$. Such uninformative labels are naturally excluded from information ratios without altering non-zero description degrees.

For numerical attributes $a \in A_n$, training values are normalized to $[0, 1]$ via min-max scaling:

$$z_a(o_i) = \begin{cases} \dfrac{a(o_i) - a_{\min}}{a_{\max} - a_{\min}}, & \text{if } a_{\max} > a_{\min}, \\ 0, & \text{if } a_{\max} = a_{\min}, \end{cases} \tag{4.2}$$

where $a_{\min}$ and $a_{\max}$ denote the minimum and maximum values of attribute a on O. Given a fixed kernel bandwidth $\tau_a > 0$, attribute-level pairwise affinities are defined as

$$K_a(i,j) = \begin{cases} \exp\left(-\dfrac{(z_a(o_i) - z_a(o_j))^2}{2\tau_a^2}\right), & \text{if } a \in A_n, \\ 1, & \text{if } a \in A_c \text{ and } a(o_i) = a(o_j), \\ 0, & \text{if } a \in A_c \text{ and } a(o_i) \neq a(o_j). \end{cases} \tag{4.3}$$

For both feature modalities, the affinity matrix satisfies $0 \leq K_a(i,j) \leq 1$, $K_a(i,j) = K_a(j,i)$, and $K_a(i,i) = 1$.

We define the local support mass $s_a(i)$, the off-diagonal support mass $t_a(i)$, and the raw local label profile $\boldsymbol{u}_a(i)$ as follows:

$$s_a(i) = \sum_{j=1}^{n} K_a(i,j), \qquad t_a(i) = \sum_{j \neq i} K_a(i,j) = s_a(i) - 1, \tag{4.4}$$

$$u_a^k(i) = \frac{\sum_{j=1}^{n} K_a(i,j)\ell_j^k}{s_a(i)}. \tag{4.5}$$

Note that $s_a(i) \geq 1$ and $t_a(i) \geq 0$, ensuring $\boldsymbol{u}_a(i) = (u_a^1(i), \ldots, u_a^q(i))$ remains a valid label distribution. For categorical attributes, $\boldsymbol{u}_a(i)$ yields the conditional mean distribution for category $a(o_i)$; for numerical attributes, it computes a Gaussian kernel-weighted mean label profile around $z_a(o_i)$.

Because the raw profile includes the self-label $\boldsymbol{\ell}(o_i)$, small or sparse neighborhoods can introduce high variance. To stabilize local profile estimates, we incorporate a shrinkage hyperparameter $\beta > 0$ to construct a regularized local label profile:

$$\eta_a(i) = \frac{t_a(i)}{t_a(i) + \beta}, \tag{4.6}$$

$$\widetilde{\boldsymbol{u}}_a(i) = \eta_a(i)\boldsymbol{u}_a(i) + (1 - \eta_a(i))\bar{\boldsymbol{\ell}}. \tag{4.7}$$

The confidence factor $\eta_a(i)$ depends entirely on the supporting mass provided by other objects. If an object is completely isolated ($t_a(i) = 0$), $\eta_a(i) = 0$, causing the regularized profile to revert to the global profile $\bar{\boldsymbol{\ell}}$. This shrinkage mechanism balances local discrimination with sample support. Hyperparameters τ_a and β are fixed strictly during training and are not re-tuned during feature subset evaluation.

4.1.2. Normalized Local KL Divergence

When the active label set satisfies $|\mathcal{K}_+| \geq 2$, we define the dataset-level information normalization factor

$$B_L = \log_2 \left(\frac{1}{\min_{k \in \mathcal{K}_+} \bar{\ell}^k} \right) > 0. \tag{4.8}$$

The constant B_L depends solely on the fixed global training label profile and remains independent of individual attributes or selected feature subsets.

Definition 4 (Local KL-Divergence Weight)**.** *For $|\mathcal{K}_+| \geq 2$, the local KL-divergence weight of attribute a at object o_i is defined as*

$$W_a(o_i) = \frac{1}{B_L} \sum_{k \in \mathcal{K}_+} \widetilde{u}_a^k(i) \log_2 \left(\frac{\widetilde{u}_a^k(i)}{\bar{\ell}^k} \right). \tag{4.9}$$

If $|\mathcal{K}_+| = 1$, we set $W_a(o_i) = 0$ for all $a \in A$ and $o_i \in O$.

The numerator in Eq. (4.9) measures the Kullback-Leibler (KL) divergence of the regularized local profile $\widetilde{\boldsymbol{u}}_a(i)$ from the global baseline $\bar{\boldsymbol{\ell}}$ [41]. A larger weight indicates greater divergence between the local label profile induced by attribute a and the global training profile. This representation statistic does not by itself establish statistical significance or predictive discrimination.

Proposition 2 (Boundedness and Zero-Weight Characterization)**.** *For any attribute $a \in A$ and object $o_i \in O$, $0 \leq W_a(o_i) \leq 1$. Furthermore, when $|\mathcal{K}_+| \geq 2$:*

$$W_a(o_i) = 0 \quad \Longleftrightarrow \quad \widetilde{\boldsymbol{u}}_a(i) = \bar{\boldsymbol{\ell}}.$$

Specifically, $W_a(o_i) = 0$ holds if o_i lacks supporting neighbors ($t_a(i) = 0$), if a is a constant attribute, or if all training instances share identical label distributions.

Proof. Let $p_k = \bar{\ell}^k > 0$ and $r_k = \widetilde{u}_a^k(i)$ for $k \in \mathcal{K}_+$. By Gibbs' inequality (or log-sum inequality), $D_{\mathrm{KL}}(\boldsymbol{r} \parallel \boldsymbol{p}) \geq 0$, with equality holding if and only if $\boldsymbol{r} = \boldsymbol{p}$. For the upper bound,

$$D_{\mathrm{KL}}(\boldsymbol{r} \parallel \boldsymbol{p}) = \sum_{k \in \mathcal{K}_+} r_k \log_2 r_k + \sum_{k \in \mathcal{K}_+} r_k \log_2 \left(\frac{1}{p_k} \right) \leq 0 + \sum_{k \in \mathcal{K}_+} r_k B_L = B_L.$$

Dividing by B_L yields $0 \leq W_a(o_i) \leq 1$. If $t_a(i) = 0$, Eq. (4.7) gives $\widetilde{\boldsymbol{u}}_a(i) = \bar{\boldsymbol{\ell}}$, implying $W_a(o_i) = 0$. For a constant attribute, $K_a(i, j) = 1$ for all i, j, which forces $\boldsymbol{u}_a(i) = \bar{\boldsymbol{\ell}}$ and thus $W_a(o_i) = 0$. □

4.1.3. Hybrid Discrepancy and Directed Aggregation

To isolate the contribution of information weighting, we employ standard symmetric feature discrepancies. For a categorical attribute value $v \in V_a$, let

$$r_a^k(v) = \frac{\sum_{i:a(o_i)=v} \ell_i^k}{|\{i : a(o_i) = v\}|} \tag{4.10}$$

denote the average description vector for value v. The hybrid discrepancy $\Delta_a(o_i, o_j)$ is defined as

$$\Delta_a(o_i, o_j) = \begin{cases} |z_a(o_i) - z_a(o_j)|, & \text{if } a \in A_n, \\ \dfrac{1}{2}\displaystyle\sum_{k=1}^{q} |r_a^k(a(o_i)) - r_a^k(a(o_j))|, & \text{if } a \in A_c. \end{cases} \tag{4.11}$$

Both numerical and categorical discrepancies lie in $[0, 1]$, vanish on the diagonal, and satisfy symmetry and the triangle inequality.

In addition to information-weighted profile discrepancy, we introduce a label-independent raw geometric discrepancy $G_a(o_i, o_j)$ to preserve topological separation for non-constant attributes that may carry zero local information weight:

$$G_a(o_i, o_j) = \begin{cases} |z_a(o_i) - z_a(o_j)|, & \text{if } a \in A_n, \\ 0, & \text{if } a \in A_c \text{ and } a(o_i) = a(o_j), \\ 1, & \text{if } a \in A_c \text{ and } a(o_i) \neq a(o_j). \end{cases} \tag{4.12}$$

Given a global mixture coefficient $\zeta \in (0, 1)$, we formulate the directed dissimilarity structure as follows.

Definition 5 (Information-Weighted Directed Dissimilarity)**.** *For $m = |A| > 0$ and any subset $P \subseteq A$, the forward and backward directed dissimilarities between o_i and o_j are defined by*

$$I_P(o_i, o_j) = \frac{1}{m} \sum_{a \in P} W_a(o_i) \Delta_a(o_i, o_j), \tag{4.13}$$

$$\overrightarrow{\rho}_P(o_i, o_j) = (1 - \zeta) I_P(o_i, o_j) + \frac{\zeta}{m} \sum_{a \in P} G_a(o_i, o_j), \tag{4.14}$$

$$\overleftarrow{\rho}_P(o_i, o_j) = \overrightarrow{\rho}_P(o_j, o_i). \tag{4.15}$$

By convention, $\overrightarrow{\rho}_\varnothing(o_i, o_j) = 0$.

Proposition 3 (Properties of the Directed Dissimilarity)**.** *For any $P \subseteq Q \subseteq A$:*

1. *Boundedness:* $0 \leq \overrightarrow{\rho}_P(o_i, o_j) \leq \frac{|P|}{m}$, *with* $\overrightarrow{\rho}_P(o_i, o_i) = 0$.
2. *Strict Positivity:* $\overrightarrow{\rho}_P(o_i, o_j) > 0$ *if objects* o_i, o_j *differ on at least one selected numerical or categorical attribute in* P.
3. *Asymmetry: The directional asymmetry is given by*

$$\overrightarrow{\rho}_P(o_i, o_j) - \overleftarrow{\rho}_P(o_i, o_j) = \frac{1-\zeta}{m} \sum_{a \in P} \left(W_a(o_i) - W_a(o_j)\right) \Delta_a(o_i, o_j).$$

   Symmetry holds if $W_a(o_i) = W_a(o_j)$ *for all* $a \in P$, *but equal label vectors* $\boldsymbol{\ell}(o_i) = \boldsymbol{\ell}(o_j)$ *are not required.*

4. *Monotonicity under Inclusion:* $\overrightarrow{\rho}_P(o_i, o_j) \leq \overrightarrow{\rho}_Q(o_i, o_j)$ *for fixed statistics and dataset normalization.*
5. *Restricted Triangle Inequality: The triangle inequality holds on triples* o_i, o_j, o_k *whenever* $W_a(o_i) = W_a(o_j) = W_a(o_k)$ *for all* $a \in P$.

Proof. Boundedness and strict positivity follow directly from $W_a(o_i) \in [0,1]$, $\Delta_a \in [0,1]$, and $G_a \in [0,1]$. For asymmetry, the geometric term $\frac{\zeta}{m} \sum G_a(o_i, o_j)$ is symmetric and cancels out. Monotonicity follows because appending attributes adds non-negative terms under a fixed denominator m. The restricted triangle inequality holds because both Δ_a and G_a satisfy metric triangle inequalities. □

Remark 1 (Failure of the General Triangle Inequality)**.** *The directed dissimilarity* $\overrightarrow{\rho}_P$ *does not generally satisfy the triangle inequality. Consider three objects with a single numerical attribute* $z_a(o_1) = 0, z_a(o_2) = 0.5, z_a(o_3) = 1.0$ *and label vectors* $\boldsymbol{\ell}(o_1) = (1,0), \boldsymbol{\ell}(o_2) = (0.5, 0.5), \boldsymbol{\ell}(o_3) = (0,1)$. *Let* $\tau_a = 0.5, \beta = 1$. *By symmetry,* $\widetilde{\boldsymbol{u}}_a(2) = \overline{\boldsymbol{\ell}} = (0.5, 0.5)$, *yielding* $W_a(o_2) = 0$, *whereas* $W_a(o_1) > 0$. *Evaluating forward distances gives:*

$$\overrightarrow{\rho}(o_1, o_3) = (1-\zeta) W_a(o_1) + \zeta,$$
$$\overrightarrow{\rho}(o_1, o_2) + \overrightarrow{\rho}(o_2, o_3) = \frac{1}{2}(1-\zeta) W_a(o_1) + \zeta.$$

Here, $\overrightarrow{\rho}(o_1, o_3) > \overrightarrow{\rho}(o_1, o_2) + \overrightarrow{\rho}(o_2, o_3)$ *for any* $\zeta \in (0,1)$, *showing that the directed dissimilarity* $\overrightarrow{\rho}_P$ *does not generally satisfy the triangle inequality.*

5. Distribution-Consistent Granular-Ball Representation

5.1. Label Distribution Consistency and Granular-Ball Dependency

Here, we formalize attribute-induced granular representations for LDL and derive an explained-variance dependency metric subject to support constraints. By explicitly decoupling intra-ball label consistency from inter-ball discrimination under minimum-support bounds, this formulation avoids trivial interpolation while preserving meaningful distributional separation.

5.1.1. Granular-Ball Representation under an Attribute Subset

Let $O = \{o_1, \ldots, o_n\}$ denote the set of training instances. For any attribute subset $P \subseteq A$, a deterministic granulation mechanism partitions O into a collection of mutually disjoint, nonempty granular balls denoted by $\mathcal{B}_P$. Because the underlying feature space is governed by the directed dissimilarity introduced in Section 4.1 rather than a standard metric, each granular ball $B \in \mathcal{B}_P$ is characterized by a triplet comprising its constituent instance subset B, a medoid representative $c_P(B)$, and a radius descriptor $r_P(B)$.

The representative $c_P(B)$ is chosen as the medoid minimizing aggregate symmetric dissimilarity:

$$c_P(B) = \operatorname*{argmin}_{o \in B} \sum_{o' \in B} d_P(o, o'), \qquad d_P(o, o') = \max\left\{ \overrightarrow{\rho}_P(o, o'), \overrightarrow{\rho}_P(o', o) \right\}, \tag{5.1}$$

where ties are deterministically resolved by the canonical index ordering of O. Although d_P does not necessarily satisfy the triangle inequality, it preserves the information-weighted directed deviations in both directions. The directional radii and the combined radius are given by

$$r_P^{\rightarrow}(B) = \max_{o \in B} \overrightarrow{\rho}_P(c_P(B), o), \qquad r_P^{\leftarrow}(B) = \max_{o \in B} \overrightarrow{\rho}_P(o, c_P(B)), \tag{5.2}$$

$$r_P(B) = \max\left\{ r_P^{\rightarrow}(B), r_P^{\leftarrow}(B) \right\}. \tag{5.3}$$

By convention, setting $P = \varnothing$ yields the trivial partition $\mathcal{B}_\varnothing = \{O\}$. Furthermore, instances that are identical across all selected attributes P are treated as an indivisible unit; label distributions alone cannot fracture attribute-indistinguishable instances.

5.1.2. Within-Ball Label Distribution Mean, Dispersion, and Consistency

Let $q = |\mathcal{Y}|$ be the number of labels, $n_B = |B|$ the ball cardinality, and $\boldsymbol{p} = \overline{\boldsymbol{\ell}} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{\ell}_i$ the global mean label distribution across the training set.

For any ball B, we define its mean label distribution $\boldsymbol{\mu}_B$ and within-ball dispersion $V(B)$ as

$$\boldsymbol{\mu}_B = \frac{1}{n_B} \sum_{o_i \in B} \boldsymbol{\ell}_i, \tag{5.4}$$

$$V(B) = \frac{1}{n_B} \sum_{o_i \in B} \|\boldsymbol{\ell}_i - \boldsymbol{\mu}_B\|_2^2, \qquad C(B) = 1 - V(B). \tag{5.5}$$

Since each $\boldsymbol{\ell}_i \in \Delta^{q-1}$ resides on the probability simplex, the center $\boldsymbol{\mu}_B$ is also a valid probability vector. The squared Euclidean loss in label space ensures an exact ANOVA-style variance decomposition.

Proposition 4 (Properties of Within-Ball Label Distribution Dispersion)**.** *For any granular ball B, the dispersion satisfies $0 \le V(B) \le 1 - 1/q$, and the consistency score satisfies $1/q \le C(B) \le 1$. Specifically:*

(i) $C(B) = 1$ (or $V(B) = 0$) if and only if all instances in B share an identical label distribution.

(ii) If all $\boldsymbol{\ell}_i$ are degenerate one-hot distributions, $V(B)$ reduces to the standard empirical Gini impurity: $1 - \sum_{k=1}^{q} (\mu_B^k)^2$.

Proof. Expanding the squared distance around $\boldsymbol{\mu}_B$ yields

$$V(B) = \frac{1}{n_B} \sum_{i \in B} \|\boldsymbol{\ell}_i\|_2^2 - \|\boldsymbol{\mu}_B\|_2^2.$$

Because $\boldsymbol{\ell}_i \in \Delta^{q-1}$, we have $\|\boldsymbol{\ell}_i\|_2^2 \le 1$. By the Cauchy–Schwarz inequality, $\|\boldsymbol{\mu}_B\|_2^2 \ge 1/q$. Hence, $0 \le V(B) \le 1 - 1/q$. The nonnegativity of squared norms implies that $V(B) = 0$ if and only if $\boldsymbol{\ell}_i = \boldsymbol{\mu}_B$ for all $o_i \in B$. When $\boldsymbol{\ell}_i$ are one-hot vectors, $\|\boldsymbol{\ell}_i\|_2^2 = 1$, which immediately yields the Gini impurity formulation. □

Proposition 4 highlights that dispersion measures distribution heterogeneity rather than informational entropy: a high-entropy but identical distribution (e.g., uniform $\boldsymbol{\ell}_i = (1/q, \ldots, 1/q)$) achieves perfect consistency ($V(B) = 0$).

To set an adaptive consistency threshold, compute the global label dispersion

$$V_0 = \frac{1}{n} \sum_{i=1}^{n} \|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2. \tag{5.6}$$

Given a minimum support $s_{\min} \geq 2$, a radius bound $\alpha \in [0, 1]$, and a relative tolerance $\kappa \in [0, 1)$, we define the relative consistency threshold as $\delta = \kappa V_0$. Both V_0 and δ are determined strictly on the training partition. A ball B is termed *eligible* if it jointly satisfies support, consistency, and compactness constraints:

$$\mathcal{E}_P = \{B \in \mathcal{B}_P : n_B \geq s_{\min},\ V(B) \leq \delta,\ r_P(B) \leq \alpha\}. \tag{5.7}$$

The condition $s_{\min} \geq 2$ explicitly excludes singletons from $\mathcal{E}_P$, preventing trivial overfitting through pure interpolation.

5.1.3. Granular-Ball Dependency Based on Explained Variance

To evaluate the representative capacity of $\mathcal{B}_P$, instances belonging to eligible balls are represented by their local centers $\boldsymbol{\mu}_B$, whereas instances in ineligible balls fall back to the global prior $\boldsymbol{p}$. The resulting training reconstruction error is

$$R(P) = \frac{1}{n}\left[\sum_{B\in\mathcal{E}_P}\sum_{i\in B}\|\boldsymbol{\ell}_i - \boldsymbol{\mu}_B\|_2^2 + \sum_{B\in\mathcal{B}_P\setminus\mathcal{E}_P}\sum_{i\in B}\|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2\right]. \tag{5.8}$$

Definition 6 (Support-Constrained Granular-Ball Dependency)**.** *The granular-ball dependency of attribute subset P is defined as the fraction of global label variance explained by the eligible balls:*

$$\Gamma(P) = \begin{cases} 1 - \dfrac{R(P)}{V_0} = \dfrac{\sum_{B\in\mathcal{E}_P} n_B\|\boldsymbol{\mu}_B - \boldsymbol{p}\|_2^2}{nV_0}, & V_0 > 0, \\ 0, & V_0 = 0. \end{cases} \tag{5.9}$$

Proposition 5 (Variance Decomposition and Boundedness)**.** *For any attribute subset $P \subseteq A$, $0 \leq \Gamma(P) \leq 1$ and $\Gamma(\varnothing) = 0$. For $V_0 > 0$, specifically:*

(i) $\Gamma(P) = 0$ if and only if either $\mathcal{E}_P = \varnothing$ or $\boldsymbol{\mu}_B = \boldsymbol{p}$ for all $B \in \mathcal{E}_P$.

(ii) $\Gamma(P) = 1$ if and only if each eligible ball has zero dispersion ($V(B) = 0$), and every object outside the eligible balls carries exactly the global prior $\boldsymbol{p}$.

Proof. Because $\sum_{i\in B}(\boldsymbol{\ell}_i - \boldsymbol{\mu}_B) = \mathbf{0}$, the cross-term vanishes, providing an orthogonal sum-of-squares decomposition for each ball B:

$$\sum_{i\in B}\|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2 = n_B V(B) + n_B\|\boldsymbol{\mu}_B - \boldsymbol{p}\|_2^2. \tag{5.10}$$

Summing over all $B \in \mathcal{B}_P$ and subtracting $nR(P)$ from nV_0 cancels the intra-ball dispersion terms for eligible balls, leaving exactly the numerator in Eq. (5.9). Since $0 \le R(P) \le V_0$, the bounds follow immediately. □

5.1.4. Nonmonotonicity under Attribute-Subset Changes

A critical property of $\Gamma(P)$ is that it does not generally satisfy subset monotonicity, i.e., $P \subseteq Q \not\Longrightarrow \Gamma(P) \le \Gamma(Q)$. Although the unconstrained between-group variation $\sum_{B\in\mathcal{B}_P} n_B \|\boldsymbol{\mu}_B - \boldsymbol{p}\|_2^2$ increases monotonically under nested partition refinements, the eligibility conditions and regenerated partitions create discrete changes:

- Regenerated partitions need not be nested; consistency and radius eligibility can change even though every accepted split respects minimum support.
- Transitioning from P to Q changes the subset-dependent dissimilarity structure, shifting representative medoids and ball boundaries nonlocally.

Consequently, attribute selection guided by $\Gamma(P)$ requires search procedures capable of handling non-monotonic objective landscapes.

5.2. Deterministic Granular-Ball Construction

To materialize the partition $\mathcal{B}_P$ deterministically, we provide a top-down, supervised binary granulation procedure.

5.2.1. Attribute-Indiscernibility Classes and Seed-Based Splitting

To preserve attribute indistinguishability, instances sharing identical feature projections on P are grouped into an *attribute-indiscernibility class* $G \subseteq B$. Each class is treated as an indivisible entity and indexed by its lowest-index instance u_G. Let $\mathcal{A}_P(B)$ denote the set of classes within B.

If $|\mathcal{A}_P(B)| \ge 2$, two distinct seed-class representatives (u, v) are identified by maximizing the mutual symmetric dissimilarity:

$$(u, v) = \underset{u_G, u_H : G \neq H}{\operatorname{argmax}} d_P(u_G, u_H), \quad \text{subject to } u < v, \tag{5.11}$$

with ties resolved by lexicographical index order. All classes in B are then assigned to the nearer seed:

$$B_1 = \bigcup_{G:\, d_P(u_G,u) \le d_P(u_G,v)} G, \qquad B_2 = B \setminus B_1. \tag{5.12}$$

Tied distances are assigned to the primary seed u.

5.2.2. Splitting and Stopping Criteria

A prospective split $(B \to B_1, B_2)$ is accepted if and only if it satisfies both structural feasibility and progressive improvement:

1. **Support admissibility:** $\min\{n_{B_1}, n_{B_2}\} \geq s_{\min}$.

2. **Progressive gain:** The candidate achieves a minimal reduction in either label dispersion or geometric radius:

$$g_V(B; B_1, B_2) > \varepsilon_V \quad \text{or} \quad g_r(B; B_1, B_2) > \varepsilon_r, \tag{5.13}$$

where the variance gain is defined as:

$$\begin{aligned} g_V(B; B_1, B_2) &= V(B) - \frac{n_{B_1} V(B_1) + n_{B_2} V(B_2)}{n_B} \\ &= \frac{n_{B_1} \|\boldsymbol{\mu}_{B_1} - \boldsymbol{\mu}_B\|_2^2 + n_{B_2} \|\boldsymbol{\mu}_{B_2} - \boldsymbol{\mu}_B\|_2^2}{n_B} \geq 0, \end{aligned} \tag{5.14}$$

and the radius contraction gain is $g_r = r_P(B) - \max\{r_P(B_1), r_P(B_2)\}$.

If a ball is eligible ($B \in \mathcal{E}_P$), contains fewer than $2s_{\min}$ instances, or fails both gain thresholds, it becomes a terminal leaf. The overall procedure is summarized in Algorithm 1; Figure 2 illustrates how the induced ball structure and its dependency can change non-monotonically when the attribute subset grows, which is the structural reason dependency gains must be treated as signed quantities.

Figure 2 illustrates non-monotone granulation on a synthetic eight-object construction: adding attribute b to subset $\{a\}$ reshapes the ball partition (objects re-assigned across balls, eligibility changes) and can *lower* the granular dependency despite distance monotonicity. Shading distinguishes eligible from ineligible ball membership.

Proposition 6 (Termination and Determinism)**.** *Algorithm 1 terminates in at most* $\max\{0, \lfloor n/s_{\min} \rfloor - 1\}$ *splits, outputting a disjoint partition of* O *that preserves attribute-indiscernibility classes.*

Proof. Each accepted split generates two disjoint subsets of size at least $s_{\min}$. Thus, the number of leaves L satisfies $Ls_{\min} \leq n$, which bounds the split operations by $\lfloor n/s_{\min} \rfloor - 1$ when a split is possible; otherwise the split count is zero. Determinism follows directly from the invariant tie-breaking rules on instance indices and the deterministic FIFO queue processing. □

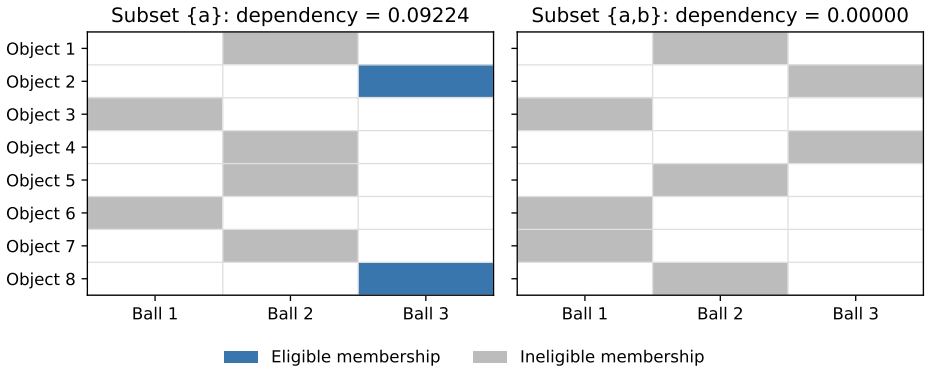


Figure 2: Non-Monotone Granular Dependency

5.2.3. Computational Complexity

With pairwise attribute distances precomputed, evaluating d_P costs $O(n^2|P|)$. For a node of size b, identifying seed-class representatives and medoids requires $O(b^2)$ operations, and computing label dispersion costs $O(bq)$. In the worst-case unbalanced tree, the cumulative granulation cost per subset evaluation is bounded by $O(n^3 + n^2 q)$. Under typical balanced splits, the time complexity scales closer to $O(n^2 \log(n/s_{\min}))$.

Algorithm 1 Deterministic Granular-Ball Partitioning for Subset P

Require: Training set O, attribute subset P, parameters $(s_{\min}, \kappa, \alpha, \varepsilon_V, \varepsilon_r)$
Ensure: Granular partition $\mathcal{B}_P$, eligible ball set $\mathcal{E}_P$, dependency $\Gamma(P)$

Compute V_0 via Eq. (5.6) and set $\delta \leftarrow \kappa V_0$
Initialize empty leaf set $\mathcal{B}_P \leftarrow \emptyset$, queue $\mathcal{Q} \leftarrow \{O\}$
while $\mathcal{Q} \neq \emptyset$ **do**
    Dequeue current member set $B \leftarrow \mathrm{pop}(\mathcal{Q})$
    if $B \in \mathcal{E}_P$ **or** $n_B < 2s_{\min}$ **then**
        $\mathcal{B}_P \leftarrow \mathcal{B}_P \cup \{B\}$; **continue**
    end if
    Form attribute-indiscernibility classes $\mathcal{A}_P(B)$
    if $|\mathcal{A}_P(B)| < 2$ **or** $\max_{G \neq H} d_P(u_G, u_H) = 0$ **then**
        $\mathcal{B}_P \leftarrow \mathcal{B}_P \cup \{B\}$; **continue**
    end if
    Find seed pair (u, v) via Eq. (5.11) and split into (B_1, B_2) via Eq. (5.12)
    if $\min\{n_{B_1}, n_{B_2}\} \geq s_{\min}$ **and** $(g_V > \varepsilon_V$ **or** $g_r > \varepsilon_r)$ **then**
        Enqueue $\mathcal{Q} \leftarrow \mathrm{push}(\mathcal{Q}, B_1, B_2)$
    else
        $\mathcal{B}_P \leftarrow \mathcal{B}_P \cup \{B\}$
    end if
end while
Extract $\mathcal{E}_P \leftarrow \{B \in \mathcal{B}_P : n_B \geq s_{\min} \wedge V(B) \leq \delta \wedge r_P(B) \leq \alpha\}$
Compute $\Gamma(P)$ via Eq. (5.9)
return $\mathcal{B}_P$, $\mathcal{E}_P$, $\Gamma(P)$

6. Objectwise-Preserving Cost-Sensitive Reduction

Figure 3 summarizes the fixed-reference search procedure of Algorithm 2. Throughout the search, the training-fitted profiles, weights, and original denominator m remain invariant, whereas each candidate subset regenerates its granular balls and anchored reconstruction. The acceptance test is consistently evaluated against the initial full-attribute error vector. A successful attribute deletion does not trigger a restart of the current sweep; instead, previously rejected attributes are reconsidered in a subsequent complete sweep. Note that the flowchart illustrates the nondegenerate-reference regime. As detailed below, the boundary case $V_0 = 0$ terminates with an `Empty` out-

put, whereas a zero full-set dependency accompanied by $V_0 > 0$ is flagged as a degenerate reference without yielding a nontrivial reduction. the final nonempty subset is guaranteed to be single-deletion irreducible, rather than representing the outcome of a global minimum-cost search.

In the flowchart, "fit and fix geometry retains the training-fitted profiles, weights and denominator m. Each candidate reconstruction regenerates the granular balls and anchored reconstruction before evaluating its errors. The sweep order is descending cost with index-based tie breaking, and its flag records whether any deletion was accepted. An unsuccessful candidate leaves P unchanged. A complete sweep with no accepted deletion yields the local stopping certificate; an empty subset terminates directly.

6.1. Anchored Extension of the Training Representation

Let $\mathcal{E}_P^0$ denote the eligible leaves generated by Algorithm 1. Their base reconstruction employs $\boldsymbol{\mu}_B$ on each eligible leaf and $\boldsymbol{p}$ elsewhere. The extension phase preserves these eligible leaves while examining deterministic alternative pairs of class representatives over the remaining portions of the base construction, subject to a fixed budget of K trials per node under identical support, compactness, and dispersion conditions. Alternative candidates are evaluated in a fixed index order, adopting the first admissible split. If no admissible pair is identified within the allocated budget, the node retains its status as a leaf.

With nonnegative within-subset safety slack ϵ_s, an additional eligible leaf B is admitted only if

$$\|\boldsymbol{\ell}_i - \boldsymbol{\mu}_B\|_2^2 \le \|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2 + \epsilon_s \quad \text{for every } i \in B. \tag{6.1}$$

Otherwise all its members retain the global fallback. Denote the resulting disjoint accepted leaves by $\widetilde{\mathcal{E}}_P$. Define

$$\widetilde{\boldsymbol{\ell}}_i(P) = \begin{cases} \boldsymbol{\mu}_B, & i \in B \in \widetilde{\mathcal{E}}_P, \\ \boldsymbol{p}, & \text{otherwise}, \end{cases} \qquad e_i(P) = \|\boldsymbol{\ell}_i - \widetilde{\boldsymbol{\ell}}_i(P)\|_2^2. \tag{6.2}$$

Proposition 7 (Within-subset safety)**.** *For a fixed subset P, the anchored extension satisfies $e_i(P) \le e_i^0(P) + \epsilon_s$ for every object, where $e_i^0(P)$ is its base reconstruction error. In exact arithmetic with $\epsilon_s = 0$, its aggregate dependency is at least the base dependency.*

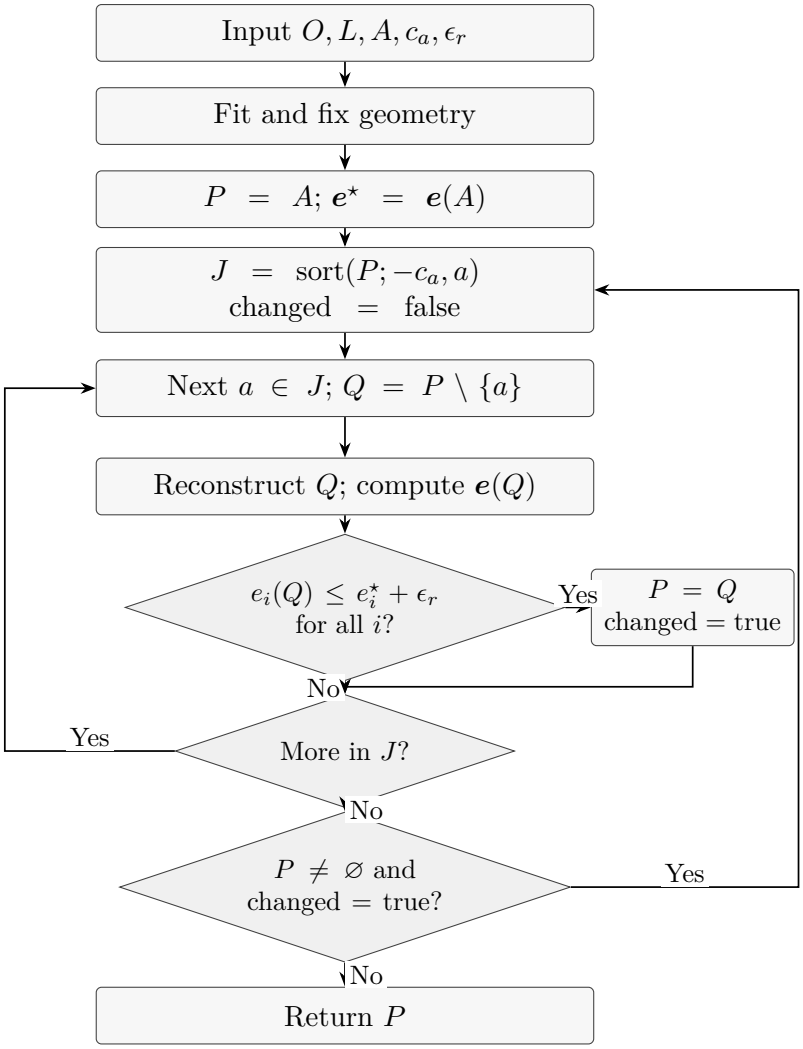


Figure 3: GBIP-O1 Workflow

Proof. Objects belonging to retained base eligible leaves preserve their original reconstructions. Any object residing in a newly accepted leaf—which previously relied on the global fallback—satisfies Eq. (6.1), whereas all remaining objects retain their fallback assignments. Averaging these individual inequalities establishes the aggregate bound via the normalized reconstruction identity whenever $V_0 > 0$. □

This is a within-subset property. Preservation across different subsets is imposed separately below.

6.2. Fixed-Reference Preservation Objective

Let $c_a = c(a) > 0$ and $C(P) = c(P) = \sum_{a \in P} c_a$. Fit all geometry statistics on the training sample, retain the original full attribute denominator m, and fix $e_i^\star = e_i(A)$. The admissible family is

$$\mathcal{F}_{\mathrm{O1}} = \{P \subseteq A : e_i(P) \le e_i^\star + \epsilon_r \text{ for all } i\}, \qquad P^\star \in \operatorname*{argmin}_{P \in \mathcal{F}_{\mathrm{O1}}} C(P). \quad (6.3)$$

The error defined in Eq. (6.2) represents the unnormalized sum of squared label-component errors. The benchmark vector $\boldsymbol{e}^\star$ is computed once prior to the search procedure and remains fixed, rather than being updated by the errors of subsequently evaluated subsets. While attribute-local profiles, information weights, and the original denominator are held constant, each candidate subset determines its specific dissimilarities and regenerates the corresponding granular balls and accepted leaves. Furthermore, the established bound controls the allowable error increase relative to the observed label distributions, rather than measuring the distance between two reconstructed distributions. For a user-defined scientific tolerance $\tau \ge 0$, the practical implementation adopts $\epsilon_r = \tau + \epsilon_{\mathrm{num}}$, where $\epsilon_{\mathrm{num}} = 10^{-12}$ accounts for numerical slack in our experiments. Consequently, strict preservation corresponds to $\tau = 0$ rather than implying a strict zero reconstruction error. Finally, kernel bandwidths τ_a and split-gain thresholds $\varepsilon_V, \varepsilon_r$ serve as distinct representation parameters, whereas ϵ_r specifically denotes the reconstruction allowance.

The full attribute set is guaranteed to be feasible. Under exact arithmetic with $\epsilon_r = 0$, any admissible subset maintains a mean reconstruction error not exceeding that of the reference baseline; however, the converse does not hold since averaging permits error compensation across instances. For $V_0 > 0$, the

tolerance-aware formulation additionally satisfies

$$\widetilde{\Gamma}(P) \geq \widetilde{\Gamma}(A) - \frac{\epsilon_r}{V_0}. \tag{6.4}$$

When $V_0 = 0$, all label distributions collapse to their global mean, rendering `Empty` an exact zero-cost representation. Conversely, if $V_0 > 0$ yet the full-set dependency is identically zero, the implementation logs a degenerate reference and terminates without yielding a nontrivial reduction. These edge cases are treated separately from the positive-reference evaluation, ensuring that no near-zero threshold replaces the specified exact-zero test.

Retaining the full feature set may stem from two distinct phenomena: either all proper subsets are infeasible, or the greedy deletion search fails to navigate toward a feasible proper subset. Consequently, rejecting all single-attribute deletions establishes solely single-deletion irreducibility; establishing the global infeasibility of compression for a given instance requires either an exhaustive search or a dedicated structural proof.

6.3. Complete-Sweep Deletion

GBIP-O1 initializes the search from the full attribute set $P = A$. Within each iteration pass, attributes present at the beginning of the pass are evaluated in descending order of their acquisition costs, with original index positions serving as a tie-breaker. For each candidate attribute a, the reduced subset $P \setminus \{a\}$ is tested against the fixed reference error vector $\boldsymbol{e}^\star$, accepting the deletion if and only if the feasibility condition in Eq. (6.3) is satisfied. Each pass runs to completion before restarting, and the search terminates when a full pass yields no accepted deletions. The complete procedure is formalized in Algorithm 2, where previously evaluated subsets can be efficiently cached via a deterministic oracle.

Proposition 8 (Feasibility, finite termination and stopping certificate)**.** *For a nondegenerate reference, GBIP-O1 maintains feasibility and makes at most $m(m+1)/2$ deletion-candidate requests. It returns either Empty or a feasible set for which every retained-attribute deletion is infeasible.*

Proof. The full attribute set is guaranteed to be feasible, and the acceptance criterion preserves this invariant. Each accepted deletion strictly reduces both subset cardinality and total cost. A successful pass starting with s attributes evaluates at most s candidates, resulting in a subset of size at most $s - 1$. Consequently, the starting cardinalities across successive passes are

Algorithm 2 GBIP-O1: Complete-Sweep Deletion from the Full Attribute Set

Require: Nondegenerate full reference, positive costs c_a, deterministic training error oracle $\boldsymbol{e}(P)$, tolerance ϵ_r

$P \leftarrow A$; $\boldsymbol{e}^\star \leftarrow \boldsymbol{e}(A)$
repeat
  changed $\leftarrow$ false; $J \leftarrow \mathrm{sort}(P; -c_a, a)$
  for each $a \in J$ **do**
    $Q \leftarrow P \setminus \{a\}$
    if $e_i(Q) \leq e_i^\star + \epsilon_r$ for every i **then**
      $P \leftarrow Q$; changed $\leftarrow$ true
    end if
  end for
until $P = \varnothing$ or changed = false
return P

bounded by $m, m-1, \ldots, 1$, including the final unsuccessful verification pass if present, yielding a cumulative upper bound of $m(m+1)/2$. In a terminating (unchanged) pass, the evaluated subset remains invariant throughout, and every deletion of a retained attribute is tested against this identical base; consequently, all such candidate deletions are proven infeasible upon termination. □

Single-deletion irreducibility constitutes a local certificate: under nonmonotone granulation, multiple simultaneous attribute deletions may well yield another valid feasible subset. In contrast, exhaustive enumeration over all 2^m subsets followed by a comparison of feasible costs constitutes a distinct global-optimality procedure, as no monotonicity-based subtree pruning is presupposed or relied upon herein.

6.4. Computational Requirements

In our reference implementation, computing the training-fitted attribute relations requires $\mathcal{O}(mn^2q)$ arithmetic operations and $\mathcal{O}(mn^2 + mnq)$ storage for numerical values. Evaluating a subset geometry incurs an aggregation cost of $\mathcal{O}(n^2|P|)$. For a fixed finite alternative-pair budget K, the conservative construction bound is $\mathcal{O}(Kn^3 + Kn^2q)$, yielding an overall deletion-search upper bound of $\mathcal{O}\Big(m^2(mn^2 + Kn^3 + Kn^2q)\Big)$ subsequent to fitting

the geometry statistics. This upper bound cleanly delineates oracle evaluations from cached implementation reuse, with independent terminal deletion verifications contributing at most m additional oracle calls. the attribute acquisition cost $C(P)$ models physical measurement expenses rather than this algorithmic computational workload.

7. Experimental Evaluation

Our empirical evaluation comprehensively assesses aggregate quality and individual instance preservation, search trajectory and geometry effects, feasibility bounds and parameter sensitivity, downstream decision consequences, and resource utilization across the image-processing pipeline.

7.1. Experimental Setup

We evaluate attribute reduction across 25 native label distribution learning (LDL) benchmark tasks originating from 16 distinct feature families, spanning scene annotation, facial expression analysis, image aesthetics, emotion recognition, gene expression, and biological aging. Table 1 summarizes the benchmark tasks and their empirical coverage. Notably, the ten Yeast labeling conditions share a common feature matrix and thus contribute a single unit of weight to family-level summaries; incomplete family–method combinations are designated as missing. Furthermore, M2B is reported separately as a predictor-suitability boundary case.

Specifically, Table 1 lists the benchmark tasks, source-array dimensions, and sample sizes per frozen split, where N, m, and q denote the numbers of source objects, attributes, and label dimensions, respectively. The training and test sizes pertain to the primary benchmark established across three independent splits per task. As noted, while the 25 main tasks draw from 16 feature families (with Yeast tasks sharing a feature space), M2B is treated independently as a dedicated boundary case.

Each dataset is partitioned into three predetermined training/test splits shared uniformly across all competing methods. Specifically, the original splits for Natural Scene, SBU-3DFE, and Yeast-cold comprise 192 training and 256 test instances. For the newly incorporated datasets, the training and test sizes are determined by $n_{\text{train}} = \min(192, \lfloor 0.75N \rfloor)$ and $n_{\text{test}} = \min(256, N - n_{\text{train}})$, respectively. Accordingly, SJAFFE utilizes 159/54 instances, Music 192/168, and Painting 192/88, whereas all remaining added

Table 1: Benchmark Datasets and Split Sizes

Task	N	m	q	Training	Test
Natural-Scene	2000	294	9	192	256
SBU-3DFE	2500	243	6	192	256
Yeast-cold	2465	24	4	192	256
Yeast-spo	2465	24	6	192	256
Yeast-alpha	2465	24	18	192	256
Yeast-cdc	2465	24	15	192	256
Yeast-diau	2465	24	7	192	256
Yeast-dtt	2465	24	4	192	256
Yeast-elu	2465	24	14	192	256
Yeast-heat	2465	24	6	192	256
Yeast-spo5	2465	24	3	192	256
Yeast-spoem	2465	24	2	192	256
FG-Net	1002	251	70	192	256
SJAFFE	213	243	6	159	54
Music	360	128	9	192	168
Painting	280	142	8	192	88
SCUT-FBP	1500	300	5	192	256
fbp5500	5500	512	5	192	256
emotion6	1980	168	7	192	256
RAF-ML	4908	200	6	192	256
Movie	7755	1869	5	192	256
Flickr	11150	168	8	192	256
Twitter	10045	168	8	192	256
MOR-PH	55132	200	62	192	256
Human-Gene	17892	36	68	192	256
M2B	1240	250	5	192	256

datasets adopt the standard 192/256 split. Every experimental run strictly operates on its designated training and test partitions.

For the primary benchmark, attribute acquisition costs are simulated as positive integers drawn from $\{1, \ldots, 10\}$ and held invariant across repeated runs of each dataset (with cost sensitivity examined later in Section 7.4.3). The original cost vectors are preserved for the three initial datasets, while the added datasets follow a uniform deterministic generation rule. During preprocessing, label vectors are normalized to sum to unity, and all finite numerical attributes—including constant columns—are retained. all predictive preprocessing steps are fitted exclusively on the respective training partition to prevent data leakage.

The cost tables report simulated budgets allocated for acquiring the selected subset of attributes. For instance, the related SCUT-FBP label distribution learning study utilizes 300 PCA coordinates extracted from image descriptors [4], with our simulation pricing each of these dimensions individually.

Furthermore, dataset names may encompass varying released versions. Specifically, the initial SCUT-FBP description reports 500 subjects [42], whereas the subsequent LDL adaptation comprises 1,500 instances [4], which precisely aligns with our array row count. Similarly, while the original LDL repository documents 30,542 human-gene objects [1], our frozen experimental source contains 17,892 instances. Because row-level transformations and identity mappings across these evolutionary versions remain unverified, all reported empirical results strictly pertain to our recorded source arrays and frozen benchmark splits.

GBIP-O1 fits its information geometry once per training split and initializes the reduction process from the full attribute set. All deletion evaluations uniformly adopt the invariant full-set reference error vector, setting the scientific tolerance to $\tau = 0$ with a numerical slack of $\epsilon_{\text{num}} = 10^{-12}$. The core representation parameters are configured as $\beta = 1$, $\zeta = 0.1$, $s_{\min} = 2$, $\kappa = 0.1$, and $\alpha = 1$, permitting at most 256 alternative seed-pair trial attempts per node. Complete cost-ordered deletion passes execute until an unchanged pass triggers termination, with each retained-attribute deletion verified independently. Predictive loss is not used in the GBIP-O1 deletion-acceptance rule; comparator parameters are selected by the training-internal procedure described below.

Comparators and Outcome Measures. The baseline comparison encompasses GBFRS, IWDLFS, and IWCLFS [9, 10]. Specifically, GBFRS-EQ denotes an equation-based reconstruction variant utilizing the disclosed neighborhood interpretation, optimized over a 200-combination grid spanning $\alpha \in \{0.05, 0.15, \ldots, 0.95\}$ and $\beta \in \{7, 9, \ldots, 45\}$. Its selected feature prefix retains the floor of 40%, 30%, or 20% of the total attributes for feature dimensions up to 100, 500, or 1,000, respectively. For the high-dimensional Movie dataset comprising 1,869 attributes, the 373-attribute output is determined via an explicitly extended 20% rule. Furthermore, the primary IW implementations restore the label-distance normalization factor prescribed in the original literature while preserving the remaining released selection logic. Each IW method selects its parameter λ from $\{0.1, 0.2, \ldots, 0.9\}$. Comprehensive reconstruction conventions and source code versions are documented alongside the implementation.

All predictive parameter choices are optimized to minimize the object-weighted Brier error across three training-internal cross-validation folds. Both feature scaling parameters and selector statistics are refitted independently within each respective fitting fold. Exact ties favor the smaller α and then the smaller β for GBFRS-EQ, and the smaller λ for each IW method. Final hyperparameter selections are fixed prior to evaluation on a separate held-out test set. This shared downstream predictor controls the prediction stage when comparing distinct attribute subsets, although the shared protocol does not reproduce each original paper's complete evaluation pipeline. Specifically, the identical training-min–max five-neighbor predictor is applied to every selected subset, leveraging stable distance tie-breaking rules and mean neighbor label distributions, whereas an `Empty` subset defaults to predicting the global training label mean. While object-weighted Brier error serves as the primary predictive metric, total variation (TV), Kullback–Leibler (KL) divergence, and Jensen–Shannon (JS) divergence are reported as supplementary measures.

GBFRS-EQ selects features without imposing the fixed objectwise constraint. In contrast, the common-preservation comparison explicitly enforces this constraint upon its feature ranking. For a matched-constraint comparison, we separately evaluate the least-cost prefix of the full GBFRS-EQ ranking that satisfies the identical objectwise reference condition as GBIP-O1, denoted as GBFRS+O1, with all evaluated prefixes verified without additional deletion passes. Furthermore, the purity-shortcut interpretation evaluated at primary GBFRS parameters, the fixed IW outputs ($\lambda = 0.8$), and

the released-distance IW variants are retained as supplementary sensitivity configurations, each reported independently.

Unavailable Outputs. The IW implementations generate nonfinite released information weights on the Painting and Movie datasets. Consequently, both methods fail to yield a valid cross-validation-selected output on these benchmarks, and their fixed-parameter configurations exhibit the same undefined status. The coverage records explicitly designate these outcomes, alongside the two degenerate GBIP-O1 cases, as unavailable. Furthermore, calculating the three-repeat means requires three valid outputs. While comprehensive supplementary tables provide every configuration, standard deviation, and operational status, M2B remains preserved as a separately identified boundary case.

Reproducibility. Frozen results, parameter settings, source manifests and verification scripts are provided in the supplementary archive. Source URLs, repository commits and checksums identify the recorded inputs and implementations. Project code and evaluation reports are available in the public GitHub repository. Frozen-record verification is distinct from rerunning feature selection, original-data prediction, image inference or hardware timing. The external inputs required for these deeper reproduction procedures are documented separately.

7.2. Aggregate Quality and Objectwise Preservation

7.2.1. Acquisition Cost and Constraint Compatibility

Table 2 summarizes the mean retained attribute cost across three frozen benchmark splits, expressed as a percentage of the full-attribute cost: $100C(P)/C(A)$, where lower values indicate lower-cost attribute subsets. Here, GBFRS+O1 denotes the least-cost feasible prefix of the GBFRS-EQ ranking evaluated under the shared evaluator. Dashes indicate unavailable three-split means, with specific reasons detailed in the protocol. Furthermore, full attribute retention is established from exact subset records rather than rounded percentage figures.

Table 2 details the resulting subset acquisition costs. Specifically, GBIP-O1 successfully yields a nondegenerate reduction output in 73 out of the 75 main dataset–repeat instances, with all 73 outputs strictly satisfying the specified objectwise condition. Two isolated cases—comprising one repeat each of the Yeast-alpha and Painting datasets—exhibit a degenerate full-set

Table 2: Retained Attribute Cost

	O1	Without O1			O1
Task	GBIP-O1	GBFRS-EQ	IWDLFS	IWCLFS	GBFRS+O1
Natural Scene	95.00	28.44	71.54	94.30	100.00
SBU-3DFE	79.17	29.16	87.21	98.45	99.93
Yeast-cold	100.00	36.22	3.15	35.43	100.00
Yeast-spo	100.00	35.17	64.57	52.49	100.00
Yeast-alpha	–	39.37	3.15	100.00	44.62
Yeast-cdc	100.00	38.06	3.15	35.43	100.00
Yeast-diau	100.00	37.53	97.90	52.49	81.10
Yeast-dtt	100.00	42.78	93.96	100.00	100.00
Yeast-elu	98.95	41.21	3.15	55.64	100.00
Yeast-heat	100.00	39.63	65.09	3.15	100.00
Yeast-spo5	100.00	37.01	32.02	35.43	100.00
Yeast-spoem	100.00	39.63	65.35	67.72	100.00
FG-Net	100.00	29.86	30.69	72.26	100.00
SJAFFE	100.00	29.81	82.33	95.36	100.00
Music	100.00	31.78	69.77	52.66	100.00
Painting	–	27.60	–	–	66.67
SCUT-FBP	55.09	29.86	0.24	99.84	100.00
fbp5500	93.99	19.35	89.08	90.98	100.00
emotion6	87.03	30.54	64.42	63.91	99.76
RAF-ML	100.00	31.12	0.34	0.34	100.00
Movie	9.88	20.34	–	–	99.67
Flickr	100.00	30.17	43.48	91.56	100.00
Twitter	99.66	27.24	46.41	63.53	100.00
MOR-PH	100.00	31.96	0.34	0.34	100.00
Human Gene	100.00	36.83	34.50	90.68	100.00

reference and are thus reported without a reduction output; consequently, complete three-repeat means are available for 23 main datasets.

Among these 23 evaluated datasets, 8 permit lower mean acquisition costs, whereas 15 retain the full-set cost entirely. The most substantial cost reduction is achieved on Movie (90.12%), followed by SCUT-FBP (44.91%) and SBU-3DFE (20.83%), with the remaining reductions ranging from 0.34% to 12.97%. These empirical findings reveal a strong dependence on the fitted information representation: redundant attribute acquisition is successfully removable on certain datasets, whereas the specified single-deletion search retains every attribute on others.

Although the originally published selectors typically acquire fewer attributes, their underlying optimization objectives differ fundamentally from the mandated objectwise requirement. Specifically, GBFRS-EQ satisfies this requirement in only 4 out of 75 main cases, IWDLFS in 8 out of 69 valid cases, and IWCLFS in 20 out of 69 valid cases. In contrast, the separately wrapped GBFRS+O1 output achieves 100% compliance, satisfying the condition across all 75 evaluated cases. Figure 4 summarizes these compliance counts at the split level rather than quantifying the number or severity of individual object-level violations, with GBFRS+O1 similarly enforcing the requirement (though its search costs are analyzed separately).

As illustrated in Figure 4, the compatibility of selected subsets with the fixed-reference objectwise bound is evaluated across 75 task–split instances. A subset is deemed *feasible* if every training object successfully passes the constraint, whereas *violating* indicates that at least one instance fails. Unavailable outputs are treated independently, comprising two degenerate GBIP-O1 references and six missing outputs per IW method due to nonfinite weights. Consequently, the feasible-to-valid counts for the plotted methods in standard order are 73/73 (GBIP-O1), 4/75 (GBFRS-EQ), 8/69 (IWDLFS), 20/69 (IWCLFS), and 75/75 (GBFRS+O1).

7.2.2. Downstream Prediction

Table 3 reports mean test Brier error across three frozen splits under the common five-neighbor predictor; lower values indicate smaller prediction error. Full uses all attributes; Empty predicts the training label mean. Dashes denote unavailable three-split means. Comparisons use unrounded values; split results and standard deviations are supplied in the supplement.

In Table 3, across the 23 datasets with complete GBIP-O1 means, its mean Brier error is lower than Full on three datasets, higher on five, and e-

Table 3: Test Brier Error

Task	Full	Empty	GBIP-O1	GBFRS-EQ	IWDLFS	IWCLFS
Natural Scene	0.24528	0.30422	0.24675	0.23767	0.25345	0.25132
SBU-3DFE	0.03115	0.03227	0.03148	0.03068	0.03137	0.03098
Yeast-cold	0.00771	0.00655	0.00771	0.00766	0.00773	0.00767
Yeast-spo	0.00956	0.00792	0.00956	0.00947	0.00950	0.00945
Yeast-alpha	0.00076	0.00064	–	0.00075	0.00076	0.00076
Yeast-cdc	0.00106	0.00092	0.00106	0.00107	0.00107	0.00106
Yeast-diau	0.00419	0.00363	0.00419	0.00414	0.00417	0.00416
Yeast-dtt	0.00340	0.00309	0.00340	0.00348	0.00343	0.00340
Yeast-elu	0.00108	0.00091	0.00109	0.00107	0.00110	0.00109
Yeast-heat	0.00506	0.00441	0.00506	0.00497	0.00507	0.00512
Yeast-spo5	0.02267	0.01879	0.02267	0.02128	0.02163	0.02163
Yeast-spoem	0.02778	0.02564	0.02778	0.02774	0.02879	0.02861
FG-Net	0.08668	0.07519	0.08668	0.08150	0.07551	0.08700
SJAFFE	0.01852	0.02854	0.01852	0.02202	0.01909	0.01881
Music	0.02612	0.02093	0.02612	0.02455	0.02527	0.02384
Painting	0.14665	0.13376	–	0.14330	–	–
SCUT-FBP	0.15590	0.13538	0.15511	0.15752	0.13871	0.15637
fbp5500	0.06766	0.16142	0.06932	0.06405	0.06683	0.06812
emotion6	0.20804	0.18777	0.20785	0.20936	0.21338	0.21285
RAF-ML	0.20525	0.24233	0.20525	0.18498	0.20816	0.20816
Movie	0.04863	0.04554	0.04387	0.04586	–	–
Flickr	0.25063	0.22598	0.25063	0.24809	0.25450	0.25066
Twitter	0.23219	0.20585	0.23220	0.22474	0.22963	0.23639
MOR-PH	0.07911	0.07586	0.07911	0.07123	0.05723	0.05723
Human Gene	0.01741	0.01473	0.01741	0.01767	0.01756	0.01742

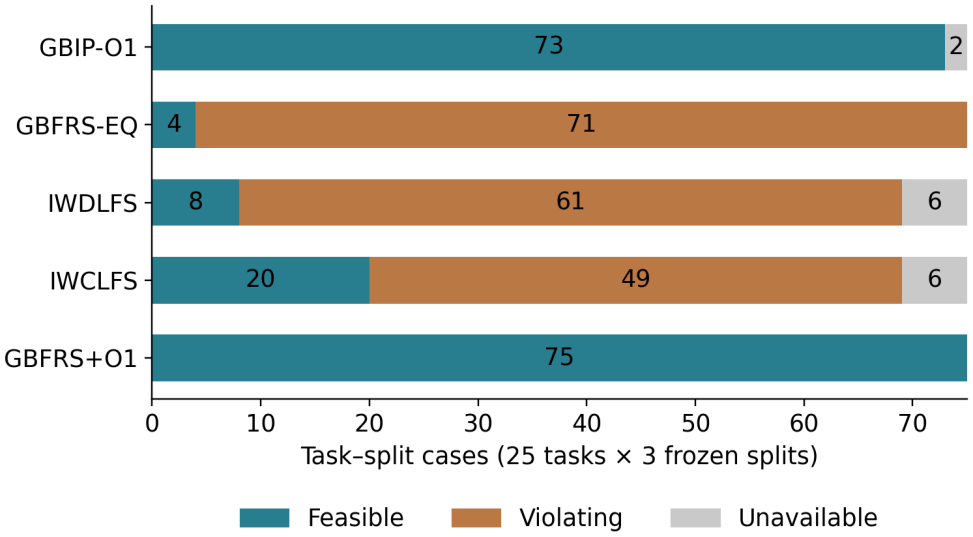


Figure 4: Objectwise Constraint Compatibility

qual on fifteen. Movie combines the largest cost saving with a decrease from 0.04862950 to 0.04386832. The remaining reduced subsets show both predictive improvements and deterioration. Figure 5 shows the cost savings and the variation in predictive changes across repeats. The fifteen cost ties correspond to retaining the full set and therefore reproduce the full-set predictions. Empty outperforms Full on several tasks, distinguishing the predictive utility of the selected attributes from training-reconstruction preservation.

GBFRS-EQ achieves both lower mean cost and lower mean held-out Brier error than GBIP-O1 on sixteen of the 23 datasets. These results do not establish a general cost or predictive advantage for GBIP-O1. The comparison measures prediction on held-out objects, while the O1 constraint applies to training reconstruction; the objectwise checks above quantify the distinct requirement imposed by O1.

Figure 5 illustrates GBIP-O1 cost savings and test Brier changes relative to Full on 23 complete tasks. Panel (a) averages $100[1 - C(P)/C(A)]$ over splits. In (b), circles give split-level $100[B(P) - B(A)]/B(A)$ and diamonds their mean; negative values improve prediction. Markers can overlap. Yeast-alpha and Painting have incomplete results and are excluded.

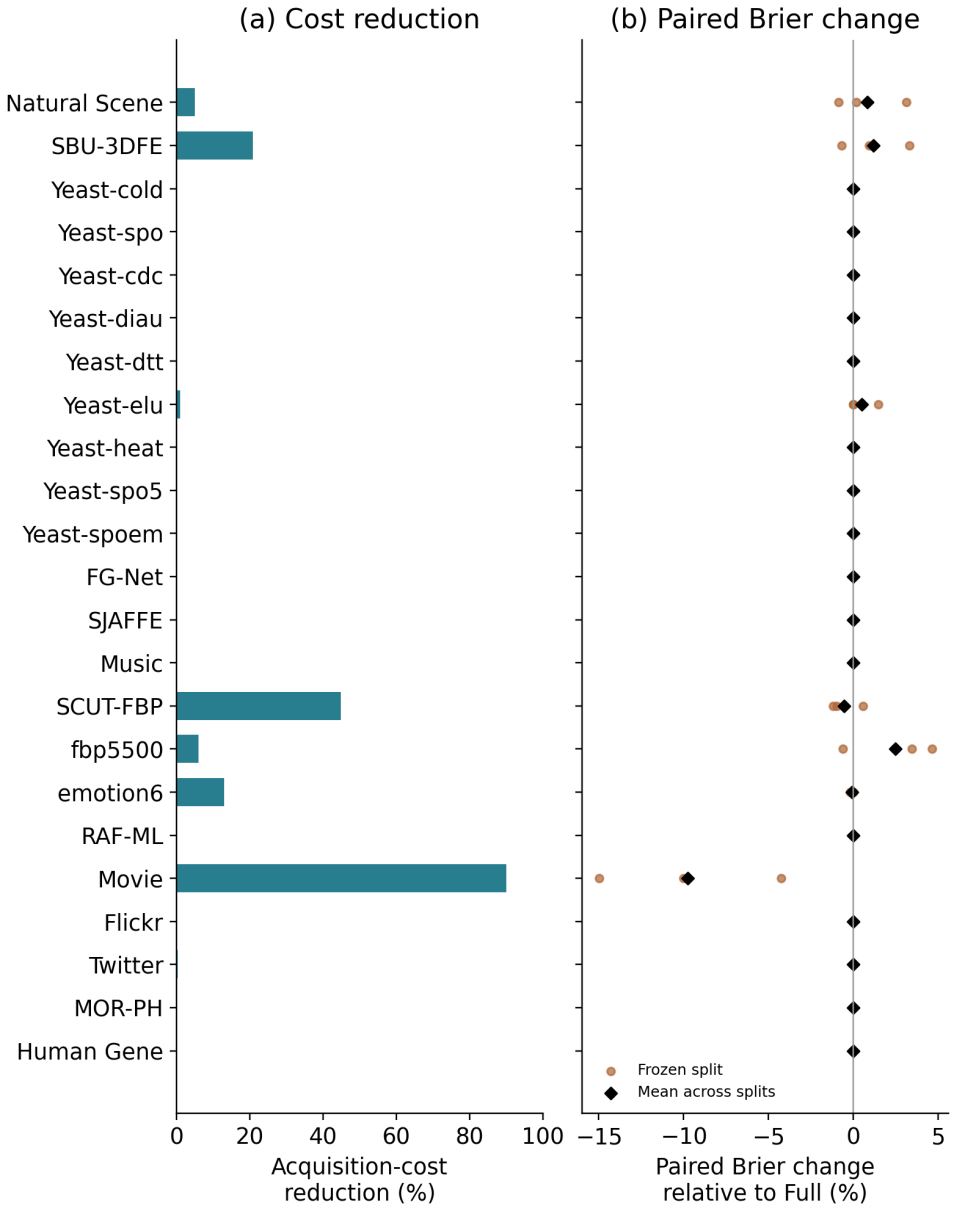


Figure 5: GBIP-O1 Cost Savings and Prediction Changes

7.2.3. Sensitivity to Predictor Feature Scaling

Qian et al. [9] evaluated selected features with AA-kNN at $k = 5$ under ten-fold cross-validation. Our unified predictor uses the same AA-kNN averaging rule [1] and specifies Euclidean distance after training-fold min–max scaling. The original GBFRS experimental description leaves the predictor input scaling unspecified. We freeze the subsets, selector parameters, costs and splits, and apply the same feature transformation to both methods before prediction. The alternatives use the released feature coordinates directly or apply training-fold z-score scaling. At matched min–max scaling, the AA-kNN calculation reproduces the unified predictor to numerical precision.

Table 4 reports the number of tasks on which GBFRS-EQ has lower cost, or both lower cost and lower Brier error, than GBIP-O1 under a shared predictor input scaling. Counts use the 23 tasks with three complete paired splits.

Table 4: Predictor Scaling Sensitivity

Input Scaling	Lower Cost	Lower Cost and Brier
Training-fold min–max	22	16
Released coordinates	22	13
Training-fold z-score	22	14

GBFRS-EQ has lower cost and Brier error on a majority of tasks in each scaling setting, with 11 tasks common to all three sets (Table 4). These comparisons isolate scaling at prediction time. They retain the parameters and subsets obtained under the main protocol, so they do not evaluate retuning or reproduce the original ten-fold experiments.

7.2.4. Paired Comparisons across Feature Families

We conduct exploratory paired comparisons at the feature-family level [43]. Each task first averages its three frozen repeats. Within a family, each method then averages the same set of tasks with complete paired results. The nine eligible Yeast conditions contribute one block, and Yeast-alpha and Painting lack complete GBIP-O1 repeats. The GBFRS-EQ comparison thus uses 23 tasks across 15 family blocks. We use two-sided Wilcoxon signed-rank tests [44] with exact sign enumeration and Holm adjustment [45] across normalized cost and Brier error. Normalized cost is expressed as a percentage of the split's full-set cost.

Table 5 reports feature-family comparisons of GBFRS-EQ with GBIP-O1 under the main min–max protocol. Differences are GBFRS-EQ minus GBIP-O1, so negative values favor GBFRS-EQ. Cost differences are in percentage points. L/T/H counts lower, tied and higher outcomes. r_{rb} is the signed rank-biserial effect. An asterisk marks Holm-adjusted $p < 0.05$.

Table 5: Feature-Family Paired Comparisons

Outcome	L/T/H	Median Δ	r_{rb}	Holm p
Normalized cost	14/0/1	−68.04	−0.983	0.000244*
Brier error	10/0/5	−0.00157	−0.533	0.0730

The normalized-cost difference meets the adjusted significance threshold, whereas the Brier difference does not (Table 5). Excluding the partially observed Yeast family gives the same significance decisions. A Friedman test [46] of Brier error compares GBIP-O1, GBFRS-EQ, Full, IWDLFS and IWCLFS on the common 14 families, after excluding Movie for unavailable IW results. The test gives $Q = 4.384$ and permutation $p = 0.3617$ (49,999 within-family permutations), so the omnibus test does not trigger post-hoc tests. The tests treat feature families as independent blocks. Their exploratory interpretation is limited to the common coverage of this benchmark.

7.2.5. Mean Preservation versus Objectwise Preservation

A training-only diagnostic changes the feasibility condition while keeping the oracle, full-set reference, costs, deletion order, complete-pass rule, tolerance and candidate-request limit fixed. The alternative accepts P when $n^{-1}\sum_i e_i(P) \le n^{-1}\sum_i e_i(A) + 10^{-12}$. The two first-repeat cases were chosen from earlier development results to cover full retention (Yeast-cold) and cost savings (SCUT-FBP). These development-selected cases illustrate the effect of changing the constraint. The table includes the original Full references. The objectwise replays reproduce the frozen selections and full-reference error vectors exactly. Actual request counts differ with the deletion paths.

Table 6 reports training reconstruction and attribute cost under mean and objectwise preservation on two first frozen splits. The column "Exceeding" counts objects satisfying $e_i(P) > e_i(A) + 10^{-12}$. Costs are simulated; Full supplies the reference.

The complete-reference mean errors are 0.00599083 and 0.12573394, respectively. In Table 6, mean preservation achieves a lower cost and lower

Table 6: Mean versus Objectwise Preservation

Task	Constraint	Cost	Attributes	Mean Training Error	Exceeding
Yeast-cold	Full	127	24	0.00599083	0/192
	Objectwise	127	24	0.00599083	0/192
	Mean	12	3	0.00591508	9/192
SCUT-FBP	Full	1690	300	0.12573394	0/192
	Objectwise	934	172	0.12244419	0/192
	Mean	2	1	0.11635585	24/192

mean reconstruction error, but permits deterioration for 9 and 24 training objects. The maximum increases are 0.00601106 and 0.18196906. GBIP-O1 produces no increases beyond the specified tolerance in these comparisons. Figure 6 shows all training-object changes, including improvements and ties. All nine Yeast-cold objects with increased error fall outside the predefined upper quartile by deviation from the global label mean.

Figure 6 illustrates objectwise changes in training reconstruction error under two preservation rules on two exploratory splits. Each point is an object; $e_i(P) - e_i(A) > 0$ denotes deterioration and a negative value improvement. Horizontal positions follow the frozen training-object order and do not represent a difficulty ranking. The panels use different vertical scales.

7.3. Search and Geometry under Preservation Requirements

7.3.1. Cost Savings under a Shared Objectwise Constraint

We compare GBIP-O1 with GBFRS+O1, the lowest-cost feasible prefix of the GBFRS-EQ ranking under the same training error reference and tolerance. Among the 73 paired splits with valid outputs for both procedures, GBIP-O1 has lower cost on 22, the same cost on 49, and higher cost on two. Both outputs satisfy the objectwise condition in every pair. The two degenerate GBIP-O1 splits are excluded. This comparison holds feasibility fixed and compares two search strategies, while Section 7.2.2 compares costs and predictions against selectors without the shared constraint.

Table 7 reports Movie cost decomposition and deletion-order diagnostic. In panel A, the first two savings are normalized by full-set cost, whereas the third is normalized by the cost of training-nonconstant attributes. In panel B, both endpoints are evaluated using the original prices. Costs and candi-

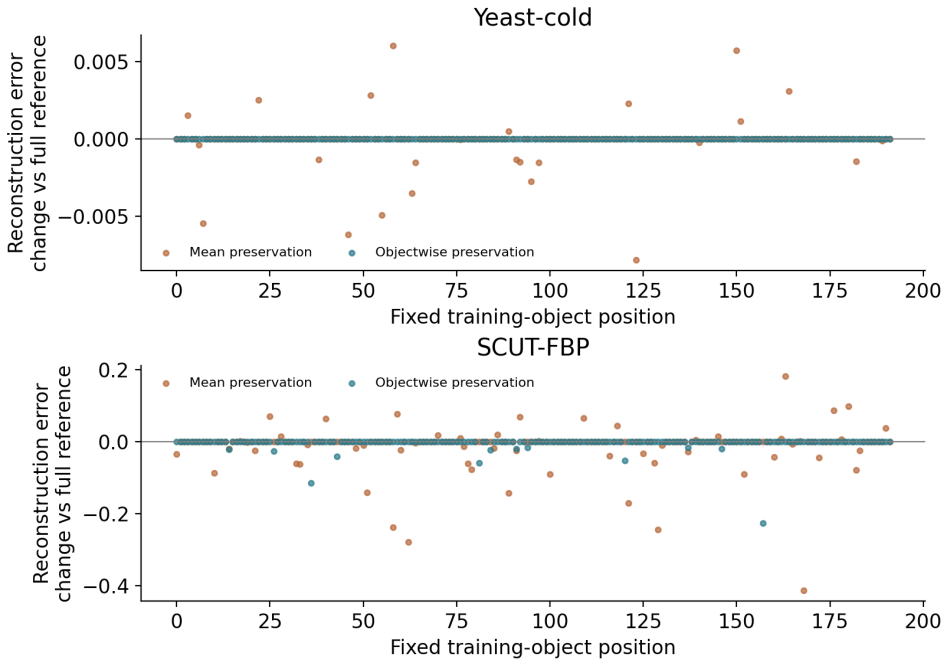


Figure 6: Objectwise Reconstruction-Error Changes

date evaluations are three-split means; evaluations exclude the independent terminal-neighborhood checks.

Movie has 1,442–1,457 columns that are constant on the respective training splits. The constant-only baseline drops these columns while keeping the original fitted reference and full-attribute denominator. Its training error vector matches Full exactly, and its test Brier matches Full on all three splits. Constant-column removal accounts for 86.10–86.64% of the cost saved by GBIP-O1, which also removes 54.06–57.62% of the cost of the remaining columns (Table 7A). In each split, 295–337 training-constant columns vary on the test set. The designation is therefore specific to the training sample. On split 1, the extra 12.43 percentage points saved beyond constant removal equal 54.06% of the remaining cost; the percentages use different denominators.

To isolate the order effect on the two largest-saving tasks, we compare descending cost with ascending feature index. The latter is the equal-cost run from Section 7.4.3, where all ordering ties are resolved by index. The two runs share the start, fitted oracle, fixed reference and complete-sweep stop-

Table 7: Movie Cost Decomposition and Deletion Order

A. Movie cost savings (%)

Split	Constant-Only Savings	Total GBIP-O1 Savings	Savings Within nonconstant Attributes
1	77.01	89.44	54.06
2	78.13	90.18	55.08
3	78.14	90.74	57.62

B. Deletion order under the same prices

	Cost Retained (%)		Candidate Evaluations	
Task	Descending Cost	Ascending Index	Descending Cost	Ascending Index
Movie	9.88	10.16	2443.3	2514.7
SCUT-FBP	55.09	47.12	1313.7	1327.0

ping rule. We evaluate both endpoints at the original prices. All six pairs have verified feasible and single-deletion-irreducible endpoints. Cost ordering gives lower retained cost on all three Movie splits and on one SCUT-FBP split. It uses fewer requests on average on both tasks, while the index order has lower mean retained cost on SCUT-FBP (Table 7B). Descending-cost deletion does not uniformly produce the lower-cost endpoint under nonmonotone feasibility.

7.3.2. Effect of Label-Information Geometry

We compare the full label-information geometry with the unweighted symmetric sum of min–max-normalized per-attribute distances, using the original dimension as denominator. Both variants use the same costs, granular reconstruction, anchored extension and complete deletion sweeps. Each fits its own full-set reference. We also score both selected subsets under the full geometry against its original full-set reference. This joint ablation removes the information weights and their associated directionality. Because each geometry defines its own reconstruction evaluator and reference, the comparison measures their combined effect on reconstruction and reduction, rather than isolating search under an identical feasible set.

On the first frozen splits, the two geometries select identical subsets on Movie and Yeast-cold, at 10.56% and 100% of full cost, respectively. The first SCUT-FBP split shows a cost difference, motivating a follow-up on its two

remaining frozen splits with the same protocol. Mean full-set reconstruction error agrees between geometries within each tested split. All selected subsets satisfy their own constraints and also pass the original full-geometry check. The reverse check fails for the full-geometry subset on SCUT-FBP split 3, where six training objects exceed the raw-geometry reference.

Table 8 reports the effect of label-information geometry on the three frozen SCUT-FBP splits. Retained cost is a percentage of full cost. Differences are unweighted symmetric minus full geometry, in percentage points; positive values favor full geometry. Each fits its own reference; all endpoints additionally pass the original full-geometry check.

Table 8: Label-Information Geometry Ablation

	Cost Retained (%)			Test Brier	
Split	Full Geometry	Unweighted Symmetric	Cost Difference	Full Geometry	Unweighted Symmetric
1	55.27	62.84	+7.57	0.16610	0.16611
2	47.87	46.69	−1.18	0.15487	0.15689
3	62.13	59.41	−2.72	0.14436	0.14603
Mean	55.09	56.31	+1.22	0.15511	0.15635

Full geometry lowers retained cost on the first SCUT-FBP split and raises it on both follow-up splits (Table 8). It makes more candidate requests in all three splits, averaging 1,314 versus 1,127, and yields slightly lower test Brier in each split. The ablation changes information weighting and directionality together.

7.4. Feasibility and Sensitivity

7.4.1. Why Do Some Searches Retain the Full Set?

For the fifteen datasets retaining full cost in all three repeats, all 4254 retained-attribute deletion checks violate the objectwise condition. Each retained attribute was tested separately at the endpoint. The number of deteriorating objects in an individual check ranges from 1 to 44 across these records.

Because the constraint is nonmonotone, single-deletion blocking does not exclude feasible simultaneous deletions. Exhausting all 276 attribute pairs on Yeast-cold repeat 1 finds no feasible pair. A further fixed-budget diagnostic covers the other seven full-retention feature families: FG-Net, SJAFFE,

Music, RAF-ML, Flickr, MOR-PH and Human-Gene. Each uses all three frozen splits and 64 unique subsets per split, fixed before oracle evaluation: 16 double deletions, 16 triple deletions, 16 subsets retaining $\lfloor m/2 \rfloor$ attributes and 16 retaining $\lfloor m/4 \rfloor$. All 21 full-reference vectors match exactly. None of the 1344 candidates is feasible; the minimum number of harmed objects over the 192 candidates per family is 3, 3, 4, 6, 9, 4 and 18, respectively.

The single-deletion checks certify local irreducibility. The joint-deletion findings apply to the specified candidates, with Yeast-cold as the only Yeast condition tested.

7.4.2. Sensitivity to Training-Sample Size

We extend three feature families using nested training samples while retaining each repeat's original 256 test objects. The original 192 training objects form the prefix; a predetermined permutation adds objects outside that training/test split. Yeast-cold uses $n = 192, 384, 768, 2209$, SCUT-FBP uses $n = 192, 384, 768, 1244$, and Human-Gene uses $n = 192, 384, 768$. The largest Yeast-cold and SCUT-FBP samples contain every object outside the fixed test set; Human-Gene remains capped at 768. All attributes, acquisition costs, representation parameters and deletion rules remain unchanged. Each size refits its geometry and full-set reference. Consequently, reference vectors differ across sizes rather than defining a common cross-size error threshold.

All 33 size–repeat cells complete, producing 99 Full, GBIP-O1 and mean-preservation evaluations. The nine 192-object references and GBIP-O1 selections reproduce the previous outputs exactly. An independent recomputation from the frozen subsets reproduces the four reported prediction metrics within 10^{-12}. The supplement documents the selection hashes and reconstruction checks.

Table 9 reports sensitivity to nested training size across three frozen repeats. Costs are percentages of full acquisition cost; GBIP-O1 reports mean $\pm$ sample standard deviation. For the Mean selector, the "Exceeding" column gives the mean percentage of objects whose errors exceed their respective references by more than 10^{-12}. Each size fits its own reference.

Table 9 shows that acquisition savings are not stable under this sample expansion. On SCUT-FBP, mean cost retained rises from 55.09% at 192 objects to 97.36% at 384, then 100% in all three repeats at both 768 and 1244. Yeast-cold and Human-Gene retain the full cost at every tested size. All nine GBIP-O1 outputs at the largest size of each family are therefore full-set outputs and reproduce Full's held-out predictions. The small SCUT-

Table 9: Training-Size Sensitivity

Task	n	O1 Cost (%)	Mean Cost (%)	Mean-Selector exceeding (%)
Yeast-cold	192	100.00 ± 0.00	7.09	7.47
Yeast-cold	384	100.00 ± 0.00	6.82	8.77
Yeast-cold	768	100.00 ± 0.00	5.77	7.99
Yeast-cold	2209	100.00 ± 0.00	32.55	9.37
SCUT-FBP	192	55.09 ± 7.13	0.36	12.67
SCUT-FBP	384	97.36 ± 2.35	5.70	13.72
SCUT-FBP	768	100.00 ± 0.00	0.85	14.28
SCUT-FBP	1244	100.00 ± 0.00	1.14	13.88
Human-Gene	192	100.00 ± 0.00	1.89	18.58
Human-Gene	384	100.00 ± 0.00	3.78	21.01
Human-Gene	768	100.00 ± 0.00	59.97	12.07

FBP predictive improvements at 192 and 384 disappear when the output becomes the full set (Figure 7). All 33 GBIP-O1 outputs satisfy the recorded objectwise condition, but this feasibility cannot be interpreted as effective reduction.

Figure 7 presents the nested training-size diagnostic. Lines connect three-repeat means; faint points show repeats. The upper panels report retained cost; the lower panels report the test Brier difference $B(P) - B(A)$ relative to Full fitted on the same sample. Negative differences indicate lower Brier error. Test objects remain fixed; each size refits its geometry and reference. Human-Gene remains capped at 768.

Mean preservation often admits substantially lower-cost subsets while allowing individual deterioration (Table 9). It is not uniformly predictable from sample size: for example, its Human-Gene cost at 768 varies strongly across repeats, as the individual points in Figure 7 show. Larger samples change the objectwise requirements, fitted geometry and reference representation together; the present design does not isolate their causal contributions. The results show that the observed cost savings can depend on training-sample size.

7.4.3. Sensitivity to Simulated Attribute Costs

We distinguish repricing a fixed subset from rerunning the cost-ordered search. Both analyses use the original data splits and express subset cost as

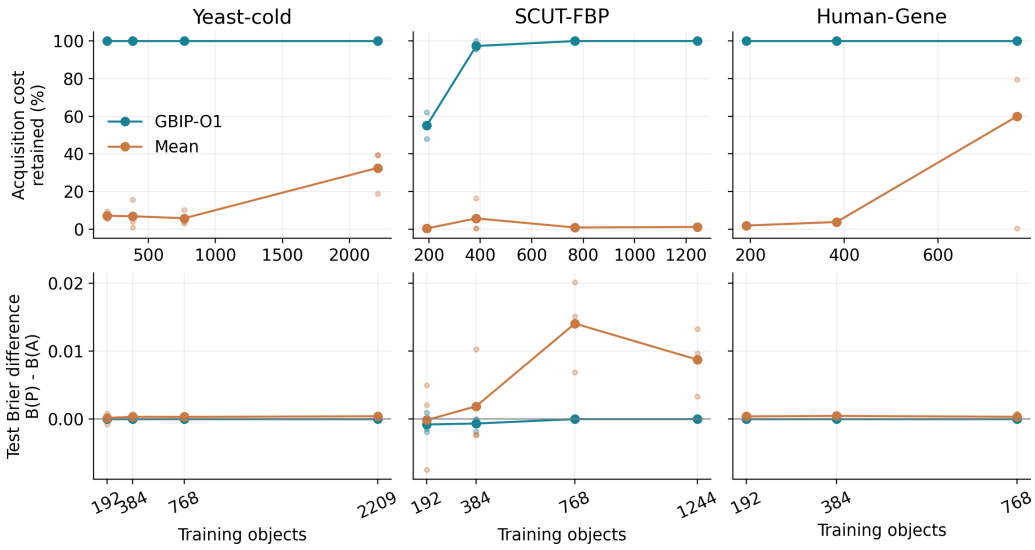


Figure 7: Cost and Prediction across Training Sizes

a fraction of the full cost under the same vector. For repricing, we apply the original costs, equal costs, two draws of uniform integers from 1 to 10, and two lognormal draws with log-scale standard deviation 1.5. The first panel of Table 10 uses the 23 tasks with three valid GBIP-O1 outputs. For fixed subsets and positive prices, the same eight tasks necessarily retain savings; the sensitivity concerns their magnitude. GBFRS-EQ is cheaper and has lower Brier error on the same sixteen tasks under all six settings. Movie and SCUT-FBP account for 72.10–77.85% of total savings across these settings.

Table 10 reports the effects of attribute-cost assumptions on fixed-subset valuation and cost-ordered search. Costs are normalized by full-set cost under the corresponding prices. Panel A reprices fixed subsets and averages over 23 tasks or 15 families. Panel B reports three-split rerun means; parenthesized values give the original subset's cost under the corresponding rerun prices. Weight-aligned/reversed costs are simulated stress tests.

We rerun GBIP-O1 on all three original splits of Movie and SCUT-FBP, the two tasks with the largest original savings. The seven alternative vectors add two stress tests to the equal and random costs. These assign the same lognormal prices in increasing or decreasing order of mean training-fitted information weight. The weight alignment is a simulated proxy for systematic cost variation. All 42 alternative searches change the selected subset and end

Table 10: Attribute-Cost Sensitivity

Fixed subsets: GBIP-O1 cost saving (%)		
Cost Setting	Task Mean	Family Mean
Original	7.88	12.02
Equal	7.70	11.73
Uniform draws 1 and 2	7.64–7.65	11.67
Lognormal draws 1 and 2	7.01–8.01	10.70–12.22
Reordered search: GBIP-O1 cost retained (%)		
Cost setting	Movie	SCUT-FBP
Original	9.88 (9.88)	55.09 (55.09)
Equal	10.01 (9.95)	46.11 (56.56)
Weight-aligned	48.16 (48.73)	65.39 (66.10)
Weight-reversed	0.47 (0.47)	48.98 (60.37)

at feasible, single-deletion-irreducible outputs. Under matched prices, 23 are cheaper than the original subset, 18 are more expensive, and one ties. The search panel shows selected settings. In Movie, the aligned and reversed vectors yield substantially different retained costs even when the original subset is held fixed. Across the benchmark, 51 splits reject every single deletion in the original first pass. Because price only changes the scan order, the single-deletion procedure returns the full set on these splits under every positive price vector, with the oracle, tolerance and reference fixed. Global exclusion of proper subsets would follow under the additional assumption that each object's error is nonincreasing as features are added: every proper subset lies inside at least one rejected single deletion and would inherit that deletion's violating object. Regenerating the reconstruction for each subset leaves that monotonicity assumption unestablished.

The accompanying M5 materials contain all split-level results and budget–preservation curves. For each budget, a curve shows the smallest observed worst-object error increase over verified candidates within that budget. The curves describe the queried candidate pool. The two-task reruns and the limited number of random draws delimit the scope of the sensitivity comparison.

7.4.4. Sensitivity to Objectwise Tolerance

We vary the additive tolerance $\tau \in \{0, 0.001, 0.01, 0.05, 0.1\}$ in the squared training reconstruction loss, with 10^{-12} numerical slack. Movie, SCUT-FBP and Yeast-cold each use their first frozen split. Each run starts from the full set with the original costs, geometry and reference. The strict setting reproduces the original search, and every endpoint passes its own tolerance and terminal single-deletion checks. Table 11 contrasts the strict setting with $\tau = 0.01$. Full grid results, worst-object increases and the sensitivity plot appear in the supplement.

Table 11 reports the effect of reconstruction tolerance on GBIP-O1 outcomes on the first frozen split of three tasks. Each tolerance starts a new full-set search under the same geometry, reference and prices. Costs are percentages of full cost. The column "Exceeding" counts objects with error increases beyond numerical slack at $\tau = 0.01$, not violations of that relaxed bound.

Table 11: Reconstruction-Tolerance Sensitivity

	Cost Retained (%)			Test Brier	
Task	Strict	$\tau = 0.01$	Exceeding	Strict	$\tau = 0.01$
Movie	10.56	7.41	9/192	0.04552	0.04782
SCUT-FBP	55.27	56.45	4/192	0.16610	0.16203
Yeast-cold	100.00	0.00	15/192	0.00721	0.00644

On SCUT-FBP, the larger feasible set leads to a higher-cost endpoint at $\tau = 0.01$, because the greedy deletion path changes. Yeast-cold retains all attributes at $\tau = 0.001$ and returns the empty set at $\tau = 0.01$, with a maximum training error increase of 0.00601. These are experimental tolerances for the stated loss. An application threshold would require a task-specific justification.

7.5. Decision Consequences

7.5.1. Reconstruction Error and Dominant-Label Changes

We reconstruct the frozen Full, GBIP-O1 and mean-preservation outputs for all nine 192-object cases in the three sample-size families, with their eighteen 384/768-object cases as supplementary diagnostics. This fixed 27-case analysis performs no new subset selection and reproduces the full error

vectors and three methods' mean reconstruction errors within 10^{-12}. Counts refer to training-object occurrences; an original object can recur across splits.

A dominant-label transition is assessed only when the target and reference have unique maxima at tolerance 10^{-12}. A change from the correct reference maximum to a unique incorrect maximum is counted separately from a predicted tie. Pairwise ordering diagnostics exclude target ties and assign a half loss to predicted ties. These measures describe the reconstructed training vectors; the primary evaluation remains the distributional error specified by the constraint.

Table 12 reports dominant-label transitions across the three 192-object repeats of each family (576 object occurrences per row). $C \to W$ denotes a correct unique reference maximum becoming uniquely incorrect; $W \to C$ denotes the reverse. No correct-to-tied transitions occur in these cases.

Table 12: Dominant-Label Transitions

	Mean Error	Mean Selector		GBIP-O1	
Family	increases	$C \to W$	$W \to C$	$C \to W$	$W \to C$
Yeast-cold	43	21	35	0	0
SCUT-FBP	73	24	28	2	13
Human-Gene	107	37	42	0	0

Among the 223 mean-selector error increases in Table 12, 69 include a correct-to-incorrect dominant-label transition, while GBIP-O1 keeps the correct maximum in those 69 occurrences. Their median error increases are 0.00182845, 0.04151158 and 0.00047333 in the three families, respectively. All 69 increases are at least 10^{-6}. The mean-selector improvements elsewhere are also retained: 105 uniquely incorrect reference maxima become correct in the nine cases. The cost premiums for GBIP-O1 over the mean selector average 92.91, 54.73 and 98.11 percentage points of full subset cost.

GBIP-O1 itself yields two correct-to-incorrect transitions on SCUT-FBP repeat 1. At source rows 659 and 470 (zero-based indices), the target and Full maxima are at component 2 (zero-based indexing) and the GBIP-O1 maximum is at component 1. Nevertheless, squared errors decrease from 0.03147797 to 0.00513887 and from 0.02681575 to 0.00406137. Thus the objectwise squared-error certificate can hold while a correct maximum changes. Pairwise-ordering deterioration also occurs: GBIP-O1 has three such occurrences at 192 objects and two at 384 objects on SCUT-FBP. Full object-level

records, reverse transitions and supplementary sizes accompany the results.

The squared-error bound can hold even when an object's dominant label or label ordering changes. Applications that require either decision to be preserved need a corresponding acceptance criterion.

Decision Stability Among O1-Feasible Occurrences. Table 13 separates empirical decision stability from the guarantee supplied by the loss bound. For Mean, the denominators are restricted to its individually O1-feasible occurrences. These conditional summaries characterize each selector's outputs on its own feasible set, rather than a matched-object comparison between columns.

Table 13 reports decision diagnostics for nine 192-object cases. Counts are training-object occurrences; objects may recur across splits. Apart from the first row, denominators use each method's O1-feasible occurrences. The unique-correct row additionally requires a unique correct reference maximum. Certified counts use feasible occurrences, not only unique-target objects. Pairwise loss uses the stated tie rule.

Table 13: Decision Diagnostics and Certificate Coverage

Diagnostic	GBIP-O1	Mean
Meets the O1 bound	1728/1728	1505/1728
Preserves the reference maximum set	1712/1728	1305/1505
Preserves a unique correct maximum	643/645	494/507
No increase in pairwise ordering loss	1725/1728	1493/1505
Unique correct maximum certified by the margin	261/1728	188/1505

A Sufficient Margin Condition. Let a be the unique largest component of the target distribution $\boldsymbol{\ell}_i$, and let $\gamma_i = \ell_{ia} - \max_{b \neq a} \ell_{ib} > 0$. If

$$e_i^\star + \epsilon_r < \gamma_i^2/2, \tag{7.1}$$

then every reconstruction that satisfies the O1 bound for object i has the same unique maximum a. For the derivation, denote the reconstructed vector by $\boldsymbol{q}$. If $q_b \geq q_a$ for some $b \neq a$, then $(\ell_{ia} - q_a) - (\ell_{ib} - q_b) \geq \gamma_i$. By Cauchy–Schwarz, $\|\boldsymbol{\ell}_i - \boldsymbol{q}\|_2^2 \geq \gamma_i^2/2$, contradicting the bound. The fixed reference also satisfies the bound and thus shares this maximum. In the nine 192-object cases, the condition certifies 261 GBIP-O1 occurrences. All retained the correct maximum. The remaining occurrences are outside this sufficient certificate and can still preserve their decisions empirically.

Table 14: SCUT-FBP5500 Reconstruction and Resource Use

Method	Attributes Retained	Exceeding	Mean Train Error	Test Brier	Test Rating MAE	MB	Backend Time (s)
Full	512	0	0.12201	0.10645	0.38116	6.7584	2.3704
Empty	0	859	0.15467	0.15055	0.55973	0	0.0042
GBIP-O1	512	0	0.12201	0.10645	0.38116	6.7584	2.3504
Mean	51	726	0.12108	0.12671	0.44148	0.6732	0.4388

7.6. Image-Pipeline Case Study

7.6.1. Objectwise Preservation and Storage on SCUT-FBP5500

The SCUT-FBP5500 benchmark provides per-image rating distributions for facial attractiveness prediction [5]. We use the official 3,300/2,200 train/test split and form each five-bin target from all 60 ratings. A frozen ImageNet-pretrained ResNet-18 [47] supplies 512 float32 coordinates per image. This setup is separate from the `fbp5500` array in the broad benchmark. Full, Empty, GBIP-O1 and Mean share the same frozen inputs, with no tuning on the official split. GBIP-O1 and Mean use the constraints in Sections 6 and 7.2.5, respectively. Held-out predictions use the same five-nearest-neighbor rule with training-fitted min–max scaling, with the training mean distribution used for Empty. Each retained coordinate requires four bytes per stored training image. Rating MAE is the mean absolute difference between target and predicted expected scores, using rating levels 1–5.

Table 14 reports training reconstruction, test prediction and descriptor storage on the official SCUT-FBP5500 split. The column "Exceeding" counts objects whose reconstruction errors exceed their reference errors by more than 10^{-12}. Storage is $4 \cdot 3300 \cdot |P|$ bytes, in decimal MB, excluding images, model, labels and auxiliary data. Backend time is the median for all 2,200 test objects.

Table 14 shows that Mean slightly improves the average training reconstruction error while increasing it for 726 of 3,300 objects (22.0%). Errors decrease for 713 objects, and the largest increase is 0.56036. The 51-coordinate model reduces raw feature storage by 90.04% and measured backend time by 81.49% relative to Full, with increases in held-out Brier error and rating MAE. Timing uses seven interleaved measurements per method after two warm-ups and covers normalization, coordinate selection, distances, sorting and aggregation. Image decoding and feature extraction are outside this timing scope.

GBIP-O1 retains all 512 coordinates, yielding the same training errors, predictions and storage as Full. Their small timing difference reflects measurement variation rather than different feature computations. Replaying the selection predicate on the saved error vectors confirms that each of the 512 single deletions violates the bound for 214 to 733 objects. We also check the 1,017 distinct nonempty subsets formed by joint deletions and queried along the Mean search. Each violates the bound for at least 658 objects. The replay verifies saved-vector digests and both methods' selection decisions. These joint candidates follow the Mean query path.

An independent follow-up fixes 512 additional candidate subsets before evaluation, using random seed 20261110 and excluding every previously queried subset. The candidate pool contains 256 double deletions, 64 triple deletions, 64 quadruple deletions and 32 deletions of each size 128, 256, 384 and 448. The geometry, reference and 10^{-12} numerical tolerance remain fixed. The full-reference replay matches exactly. None of the 512 candidates is feasible, either at numerical tolerance or at the recorded zero threshold (Table 15). The smallest number of violating objects is 415/3300. The smallest maximum error increase across candidates is 0.50206; its candidate violates the bound for 450 objects. The minimum violating count and minimum maximum-increase summaries need not identify the same candidate.

Table 15 reports feasibility of 512 independently specified joint-deletion candidates on SCUT-FBP5500. Candidates were fixed before evaluation and exclude previous queries. Violating objects exceed their frozen reference by 10^{-12}; min–max gives object-count ranges within each deletion-size group, not error magnitudes.

Table 15: Independent Joint-Deletion Feasibility

Attributes Deleted	Attributes Retained	Candidates	O1-Feasible Candidates	Violating Objects min–Max
2	510	256	0	415–731
3	509	64	0	517–738
4	508	64	0	563–736
128	384	32	0	695–746
256	256	32	0	694–743
384	128	32	0	694–739
448	64	32	0	708–750

Together, the two query pools cover 2,041 distinct nonempty proper subsets, all infeasible under the fixed bound. Subsets outside these pools remain unresolved. The follow-up audit checks the returned summaries against the frozen protocol and original query manifest; the candidate error vectors remain in the run's SQLite cache.

A follow-up measures the complete query pipeline from the original test images to the predicted distributions (Table 16). It uses CPU inference with four PyTorch threads, batches of 32 and the same frozen training features. Full and Mean are interleaved for two warm-up rounds and seven measured rounds each. Reading the ZIP and decoding images are included, followed by preprocessing, ResNet-18 inference and the original prediction backend. Model loading precedes measurement; repeated passes use warmed file-system caches. The total median time falls from 32.90014 to 30.97551 seconds, a 5.85% reduction. The backend median falls by 79.35%, while the shared image pipeline dominates the total. Re-extracted test features differ from their frozen values by at most 5.25×10^{-6}, and predictions match the frozen predictions exactly in every recorded round.

Table 16 reports end-to-end CPU query time for 2,200 SCUT-FBP5500 test images, in seconds. Entries are separate medians of seven interleaved rounds after two warm-ups; four PyTorch threads and batches of 32 are used. ZIP reading through prediction is included; model loading is excluded and file caches are warm. Stage medians need not sum to the total, which includes concatenation and other overhead.

Table 16: End-To-End CPU Query Time

Stage	Full	Mean
Image reading and decoding	1.79564	1.79022
Preprocessing and ResNet-18	28.81379	28.70898
Prediction backend	2.30858	0.47674
Total	32.90014	30.97551

On this split, the storage reduction comes from accepting a bound on average reconstruction error. The objectwise constraint retains the full feature matrix.

Recorded-Pool Storage Envelope on SCUT-FBP5500. We pool the 1,531 original queries and 512 independent candidates from Section 7.6.1,

including Full and Empty. For each of these 2,043 subsets, define $d(P) = \max\{0, \max_i[e_i(P) - e_i^\star]\}$. With $S(P) = 4 \cdot 3300 \cdot |P|$ bytes, the recorded-pool envelope is

$$S_{\mathcal{C}}(\tau) = \min_{P \in \mathcal{C}:\, d(P) \le \tau} S(P). \tag{7.2}$$

Numerical comparisons retain the original 10^{-12} slack. Candidate ties use the lexicographic order of sorted feature indices. The fixed pool yields a nonincreasing storage envelope; the separate greedy reruns above can change paths as tolerance changes. The six endpoints are selected using training reconstruction and storage. We then reconstruct these endpoints and reproduce all six recorded training-error-vector hashes before assessing their decisions and held-out predictions.

Table 17 reports minimum-storage endpoints within the recorded candidate pool and their decision and prediction outcomes. L_1 counts losses among 2,159 reference-correct unique top-1 decisions; L_2 among 1,973 eligible reference-correct top-2 sets. R counts ordering-loss increases among 3,300 objects. The "Certified column applies the reference-plus-tolerance bound. Tolerance knots are rounded to five decimal places.

Table 17: Recorded-Pool Storage Endpoints

τ Knot	Dimensions	MB	L_1	L_2	R	Certified	Test Brier
0.00000	512	6.7584	0	0	0	1479	0.10645
0.48122	511	6.7452	75	92	214	0	0.10635
0.50206	510	6.7320	123	139	392	0	0.10621
0.52498	89	1.1748	184	204	571	0	0.11582
0.53036	50	0.6600	192	224	583	0	0.12720
0.56036	0	0.0000	237	242	760	0	0.15055

The decision denominators are 2,159 training objects with a unique correct reference maximum, 1,973 with a correct top-2 target set and unambiguous target and reference cutoffs, and 3,300 with at least one unequal target-label pair. A predicted tie counts as loss of an unambiguous decision. The certificate column uses Eq. (7.1) with the endpoint's allowed tolerance. Top-2 outcomes are assessed directly from the reconstructed distributions.

No recorded subset reduces storage at $\tau \le 0.48$. The first reduction retains 511 coordinates at $d(P) \simeq 0.48122$, saving 0.20%. Storage falls to 89

coordinates at approximately 0.52498 and 50 at 0.53036. The latter saves 90.23% of stored-feature bytes, with 192 losses of a reference-correct top-1 decision, 224 losses of a reference-correct top-2 set, and 583 objects with increased ordering loss. At this endpoint, 197 initially incorrect unique reference maxima become correct and 557 objects have reduced ordering loss. The 50-coordinate endpoint differs from the 51-coordinate Mean endpoint measured in Table 16. The empty subset becomes admissible at 0.56036. These thresholds describe sensitivity within the recorded pool and leave the acceptance tolerance to be specified for an application.

Figure 8 illustrates storage–reconstruction trade-offs within a frozen pool of 2,043 subsets, including Full and Empty. Panels (a)–(c) assess the lowest-storage recorded feasible subset at each tolerance. Panel (b) uses metric-specific eligible denominators. Panel (d) shows sufficient-certificate coverage under the permitted error bound, not observed accuracy; it may change with the same selected subset. Test outcomes do not select endpoints.

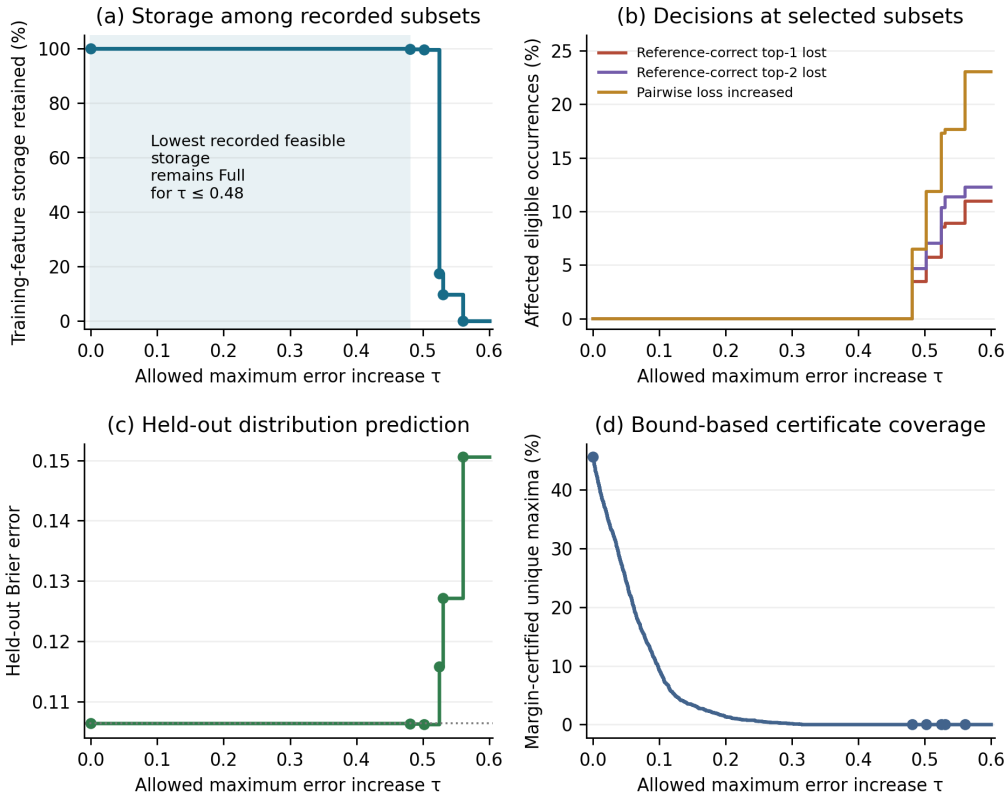


Figure 8: Storage–Reconstruction Trade-Offs

8. Discussion

The framework separates aggregate performance from individual reconstruction feasibility. GBFRS-EQ has lower acquisition cost and lower test Brier error on most tasks with complete results, whereas matched-reference comparisons isolate the costs of alternative feasible-subset searches (Sections 7.2.2 and 7.3.1). Family-level tests support a cost difference but not a Brier difference at the adjusted threshold (Section 7.2.4). These results support evaluating the specified requirement, rather than a general predictive advantage.

Full retention is common. Single-deletion checks establish local irreducibility, while joint diagnostics cover only the queried subsets. Movie's savings largely remove training-constant columns; the larger-sample experiments eliminate SCUT-FBP's initial savings. Prices affect both subset valuation and deletion order (Section 7.4.3). Thus savings depend on the training representation and acquisition assumptions.

A larger tolerance expands the feasible set without guaranteeing a lower-cost greedy endpoint (Section 7.4.4). Geometry likewise changes costs in either direction on SCUT-FBP, despite slightly lower test Brier under full geometry (Section 7.3.2). In the image case, substantial recorded-pool compression requires appreciable worst-object deterioration and changes training decisions in both directions (Table 17). A reconstruction bound alone does not preserve label decisions [1, 4, 8]; the margin certificate and direct diagnostics distinguish these outcomes.

Resource measures also differ: Mean's 90.04% descriptor-storage reduction yields only a 5.85% end-to-end query-time reduction because feature extraction remains unchanged (Section 7.6.1). The broad benchmark instead uses simulated attribute costs, three development-consulted splits and equation-based baseline reconstructions. Independent evaluation with measured costs and original baseline protocols would extend the evidence.

Practical use requires a task-appropriate reference, loss and tolerance. The experiments quantify specified reconstruction requirements without establishing an application acceptance threshold. Held-out prediction and full-pipeline timing provide complementary measures of their consequences.

9. Conclusion

This study formulates cost-sensitive attribute reduction for label distribution learning under fixed-reference objectwise reconstruction bounds. GBIP-

O1 combines label-information geometry and granular-ball reconstruction with complete cost-ordered deletion sweeps. Each accepted deletion maintains feasibility, and termination certifies that no retained attribute can be individually removed without violating the specified constraint. A label-margin condition further provides a sufficient guarantee for preserving a unique dominant label.

Experiments on 25 tasks spanning 16 feature families, together with a separate image-pipeline study, demonstrate individual training-reconstruction deterioration under mean preservation and characterize the acquisition cost of enforcing objectwise bounds. Matched-constraint comparisons show that search strategy can affect retained acquisition cost, while geometry, training-sample size and tolerance alter the reduction outcomes. The specified deletion search can retain the full set under strict preservation; relaxed bounds quantify a storage–reconstruction trade-off. These findings support evaluating explicitly defined individual reconstruction requirements while assessing held-out prediction and measured resource use separately.

Notation and abbreviations

The tables below collect the principal mathematical symbols and abbreviations used in the paper. They supplement the definitions at first use. In particular, reconstruction allowance ϵ_r differs from the representation split threshold ε_r, and scientific tolerance τ differs from the attribute-specific bandwidth τ_a.

Table 18: Principal Mathematical Notation

Symbol	Meaning
O, o_i, n	Training-Object Set, Its iTh Object, and Its Size.
A, m, P, Q	Full Attribute Set, $m = \|A\|$, Current Subset, and Candidate Subset. the Denominator m Stays Fixed During Search.
A_n, A_c	Numerical and Categorical Attribute Sets.
$\mathcal{Y}$, q	Label Set and Number of Labels.
L, ℓ_i^k, $\boldsymbol{\ell}_i$	Label-Distribution Matrix, Description Degree of Label k for Object i, and Its Distribution Vector.
$c_a = c(a)$, $C(P) = c(P)$	Positive Acquisition Cost of Attribute a, and Additive Subset Cost $\sum_{a\in P} c_a$.
$\boldsymbol{p}$, $\bar{\boldsymbol{\ell}}$	Equivalent Global Mean Training-Label Distributions, Also Used as the Fallback Reconstruction.
$z_a(o_i)$, $K_a(i,j)$	Training-Scaled Numerical Value and Attribute-Level Affinity.
τ_a, β	Attribute-Specific Kernel Bandwidth and Local-Profile Shrinkage Parameter.
$\boldsymbol{u}_a(i)$, $\widetilde{\boldsymbol{u}}_a(i)$	Raw and Regularized Attribute-Induced Local Label Profiles.
$W_a(o_i)$, B_L	Normalized Local KL-Divergence Weight and Its Dataset-Level Normalizer.
Δ_a, G_a, ζ	Profile-Based Discrepancy, Raw Geometric Discrepancy, and Their Mixture Coefficient.
$\overrightarrow{\rho}_P$, $\overleftarrow{\rho}_P$	Forward and Backward Mixed Directed Dissimilarities for Subset P.

Table 18 (continued)

Symbol	Meaning
B, $\boldsymbol{\mu}_B$, $V(B)$	Granular Ball, Its Mean Label Distribution, and Within-Ball Label Dispersion.
V_0, $s_{\min}$, κ, α	Global Label Dispersion, Minimum Ball Support, Relative Dispersion Tolerance, and Radius Bound.
ε_V, ε_r	Representation Split-Gain Thresholds for Label Dispersion and Radius.
$\Gamma(P)$, $\widetilde{\Gamma}(P)$	Base and Anchored-Extension Aggregate Reconstruction Dependencies.
K, ϵ_s	Alternative Class-Representative-Pair Budget Per Node and Within-Subset Safety Slack.
$\widetilde{\boldsymbol{\ell}}_i(P)$, $e_i(P)$	Anchored Reconstruction and Its Unnormalized Squared Error Relative to the Observed Distribution.
$e_i^\star = e_i(A)$	Fixed Full-Attribute Reconstruction Error for Object i.
τ, ϵ_{num}, ϵ_r	Scientific Reconstruction Tolerance, Numerical Slack, and Total Allowance $\epsilon_r = \tau + \epsilon_{\text{num}}$.
$\mathcal{F}_{\text{O1}}$, $P^\star$	Feasible Subset Family and a Global Cost Minimizer in That Family; GBIP-O1 Does Not Certify Global Optimality.
J, changed	Ordered Attributes Present At the Start of a Sweep, and Its Accepted-Deletion Flag.
γ_i	Gap between the Largest and Second-Largest Target-Label Degrees for an Object with a Unique Maximum.
$B(P)$	Held-Out Test Brier Error for Subset P; Distinct from a Granular Ball B.
$\mathcal{C}$, $d(P)$, $S_{\mathcal{C}}(\tau)$	Recorded Candidate Pool, Largest Nonnegative Reference-Error Increase, and Its Minimum-Storage Envelope.

Table 19: Abbreviations and Method Labels

Term	Meaning or usage
LDL	Label Distribution Learning.
O1	the Fixed-Reference Objectwise Preservation Condition Used in This Paper.
GBIP-O1	the Proposed Complete-Sweep, Cost-Ordered Deletion Method under O1.
GBFRS	the Granular-Ball Fuzzy Rough-Set Selector of Qian Et Al. [9].
GBFRS-EQ	This Study'S Equation-Based GBFRS Reconstruction under Its Disclosed Protocol.
GBFRS+O1	Least-Cost Feasible Prefix of the GBFRS-EQ Ranking under the Shared O1 Evaluator.
IWDLFS / IWCLFS	the Dependence-Based and Conditional-Discrimination-Based Informational Weighted Selectors Evaluated Here [10].
Mean	Comparator Using Average Training-Error Preservation.
Full, Empty	All Attributes, or No Attributes; Empty Predicts the Training-Label Mean.
AA-KNN	Algorithm-Adaptation k-Nearest-Neighbor LDL Predictor.
CV, PCA	Cross-Validation; Principal Component Analysis.
KL	Kullback–Leibler Divergence.
MAE, RMSE	Mean Absolute Error; Root Mean Squared Error.
CPU, MB	Central Processing Unit; Decimal Megabyte (10^6 Bytes).

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